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Risk Prediction in Chronic Kidney Disease

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CHAPTER OUTLINE

KEY POINTS

- Because of a large variance in chronic kidney disease (CKD) progression, there is a need to develop methods for risk stratification within CKD to identify the relatively small subgroup of patients who are at risk of progression to end-stage kidney disease (ESKD) and who may benefit from intervention.
- The combination of risk factors (age, gender, ethnicity, etc.) and biomarkers (proteinuria, anemia, fibroblast growth factor 23, etc.) can be useful in risk stratification of patients with CKD.
- In a similar manner to the Framingham risk score for prediction of cardiovascular risk, renal risk scores such as the kidney failure risk equation (KFRE) have allowed for the accurate prediction of CKD progression.
- Equations such as the KFRE are externally validated and ready to be implemented in a clinical setting.

Chronic kidney disease (CKD) currently affects more than 26 million Americans[1] and a meta-analysis of studies from around the globe has reported a prevalence for CKD stages 1–5 among adults of 8.7% to 18.4% with a global mean prevalence of 13.4% [95% confidence interval (95% CI), 11.7% to 15.1%].[2] The large number of people known to be affected by CKD has major implications for the provision of health care – in particular, nephrology services. In the past two decades, nephrology has moved from a position where it provided highly specialized services to a relatively small number of patients with specific and relatively rare kidney disease or advanced CKD, to one where it must concern itself with the care of less-advanced CKD in a substantial proportion of the general population. Furthermore, early stage CKD is largely asymptomatic, and detection therefore requires a screening process. Studies have indicated that screening whole populations is not cost-effective,[3] and a means of identifying high-risk subgroups for targeted screening is therefore required. Successful screening programs are likely to identify large numbers of patients with previously undiagnosed CKD; however, in most countries, nephrology services are unable to provide long-term care to all CKD patients, and the associated costs would be prohibitive. A solution to this problem was suggested by studies showing that there is substantial heterogeneity among patients who meet the diagnostic criteria for CKD, with most being at relatively low risk of ever progressing to end-stage kidney disease (ESKD). One reason for this heterogeneity is the use of an estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73 m^2 as the standard of CKD diagnosis in clinical practice.[4] This provides susceptibility to both overdiagnosis and underdiagnosis of CKD as any given level of eGFR will have a large variation in the risk for progression to kidney failure.[5] As a result, there will be situations in which low-risk patients are overrecognized and receive unnecessary treatment or, conversely, high-risk patients being under-recognized and receiving less-than-optimal treatment with regard to CKD progression.[6] The Kidney Disease Outcomes and Quality Initiative (K/DOQI) classification system for CKD was widely adopted and proved valuable, particularly for identifying the prevalence of different stages of CKD in epidemiologic studies.[7] It was noted, however, that the classification provided little information on the future risk of decline in renal function.[8] The Kidney Disease Improving Global Outcomes (KDIGO) classification system therefore modified the K/DOQI system so that categories defined by eGFR and albuminuria do correlate with risk,[9] but this does not provide accurate, individual risk prediction. Previous studies identified a wide range of rates of decline in GFR among patients with CKD, and up to 15% may even show an increase over time.[10]

In a retrospective cohort study done by O'Hare et al. using patients with advanced CKD from the United States Renal Data System and Veterans Affairs databases, the following trajectories were discovered: 1. consistently low eGFR < 30 mL/min/1.73 m^2 with a mean yearly decrease of 7.8 mL/min/1.73 m^2, 2. progressive loss of initial eGFR at stage 3 CKD (eGFR of 30–59 mL/min/1.73 m^2) with a mean yearly decrease of 16.3 mL/min/1.73 m^2, 3. accelerated loss of initial eGFR > 60 mL/min/1.73 m^2 with a mean yearly decrease of 32.3 mL/min/1.73 m^2, and 4. dramatic loss of initial eGFR > 60 mL/min/1.73 m^2 in 6 months or less, with a mean yearly decrease of 50.7 mL/min/1.73 m^2.[11] In a similar study, Li et al. also found distinct patterns of CKD progression, this time using patients from the African American Study of Kidney Disease and Hypertension (AASK), which included 846 patients that were followed over a 3-year period. The patterns of progression included 1. stable or increasing eGFR, defined as less than 2 mL/min/1.73 m^2 decrease per year, 2. rapid decrease in eGFR, defined as a loss of more than 4 mL/min/1.73 m^2 per year, and 3. alternating combinations of the previously mentioned categories. It was determined that a stable or increasing eGFR was the most common pattern of CKD progression, with 58% of participants following such a trajectory.[12] Due to such a variance in progression of CKD, there is a need to develop methods for risk stratification within CKD to identify the relatively small subgroup of patients who are at risk of progression to ESKD and who may benefit from specialist intervention to slow or halt disease progression. Such risk stratification would be equally important for identifying individuals who are at low risk for progression who could be reassured and spared unnecessary referral to a nephrologist.

Another important aspect of CKD is its association with a substantially increased risk of future cardiovascular events (CVEs) that in most patients with mild CKD substantially exceeds the risk of ESKD.[13] Whereas CKD is associated with a high prevalence of many traditional risk factors for cardiovascular disease (CVD), such as hypertension and dyslipidemia, risk prediction tools such as the Framingham risk score substantially underestimate cardiovascular risk in patients with CKD.[14] It has been proposed that this observation is due to the role of several nontraditional cardiovascular risk factors that are specific to CKD.

From the aforementioned discussion, it is clear that there is a need to identify and understand factors associated with an increased risk of developing CKD and, once diagnosed, factors associated with an increased risk of progression to ESKD and CVE. In this chapter, we will review current knowledge of these risk factors and the methods being applied to predict risk in CKD patients. Risk factors for CVD in patients with CKD, many of which overlap with risk factors for CKD progression, are discussed in Chapter 54.

DEFINITION OF A RISK FACTOR

A risk factor is a variable that has a causal association with a disease or disease process such that the presence of said variable in an individual is associated with an increased risk of the disease being present or developing in the future. Thus risk factors are a useful tool for identifying subjects with increased risk for a disease or particular outcome due to a disease process. In the course of epidemiologic research, many variables may show associations with a disease of interest but these may be chance associations, noncausal associations, or causal associations (true risk factors). The Bradford Hill criteria provide minimum requirements to be fulfilled to identify a causal relationship between a putative risk factor (exposure) and a disease (outcome; Table 20.1). In complex diseases such as CKD that result from the combined effects of multiple factors, it is likely that many risk factors will not fulfill all the criteria. Nevertheless, they do provide a useful framework for assessing the strength of a proposed causal relationship between risk factor and disease.

RISK FACTORS AND MECHANISMS OF CHRONIC KIDNEY DISEASE PROGRESSION

It has been appreciated for several decades that once GFR has decreased to below a critical level, it tends to progress relentlessly toward ESKD. This observation suggests that loss of a critical number of nephrons provokes a vicious cycle of further nephron loss. Detailed studies have elucidated a number of interrelated mechanisms that together contribute to CKD progression, including glomerular hemodynamic responses to nephron loss [raised glomerular capillary

Table 20.1 Bradford Hill Criteria of Causality

Parameter	Explanation
Strength of association	The stronger the association, the more likely the relationship is causal.
Consistency	A causal association is consistent when replicated in different populations and studies.
Specificity	A single putative cause produces a single effect.
Temporality	Exposure precedes outcome (i.e., risk factor predates disease).
Biological gradient	Increasing exposure to risk factor increases risk of disease, and reduction in exposure reduces risk.
Plausibility	The observed association is consistent with biologic mechanisms of disease processes.
Coherence	The observed association is compatible with existing theory and knowledge in a given field.
Experimental evidence	The factor under investigation is amenable to modification by an appropriate experimental approach.
Analogy	An established cause-and-effect relationship exists for a similar exposure or disease.

Modified from Hill AB. The environment and disease: association or causation? *Proc R Soc Med.* 1965;58: 295–300.

hydraulic pressure and single-nephron GFR (SNGFR)], proteinuria, and proinflammatory responses. A generally good prognosis after unilateral nephrectomy[15] attests to the fact that a single pathogenic factor may be insufficient to initiate progressive CKD, but the multihit hypothesis proposes that multiple factors interact to overcome renal reserve and provoke progressive nephron loss.[16] To meet the Bradford Hill criteria of plausibility and coherence, a putative risk factor should therefore somehow affect known mechanisms of CKD progression (see Chapter 51 for further details). Fig. 20.1 shows how risk factors may interact with pathophysiologic mechanisms to initiate or accelerate CKD progression.

Based on our understanding of the mechanisms underlying the pathogenesis of CKD and its progression, risk factors may be divided into susceptibility factors, initiation factors, and progression factors (Table 20.2). First, susceptibility includes risk factors associated with an increased risk of an individual developing CKD after exposure to a factor that has potential to cause renal damage. An example is a reduced nephron number after uninephrectomy, which is associated with an increased risk of developing diabetic nephropathy if the individual develops diabetes.[17] Initiation factors directly cause or initiate kidney damage in a susceptible individual. Examples include exposure to nephrotoxic drugs, urinary tract obstruction, or primary glomerulopathies that may provoke CKD in some (but not all) exposed individuals. Finally, progression factors are those that contribute to the progression of kidney damage once CKD has developed. An

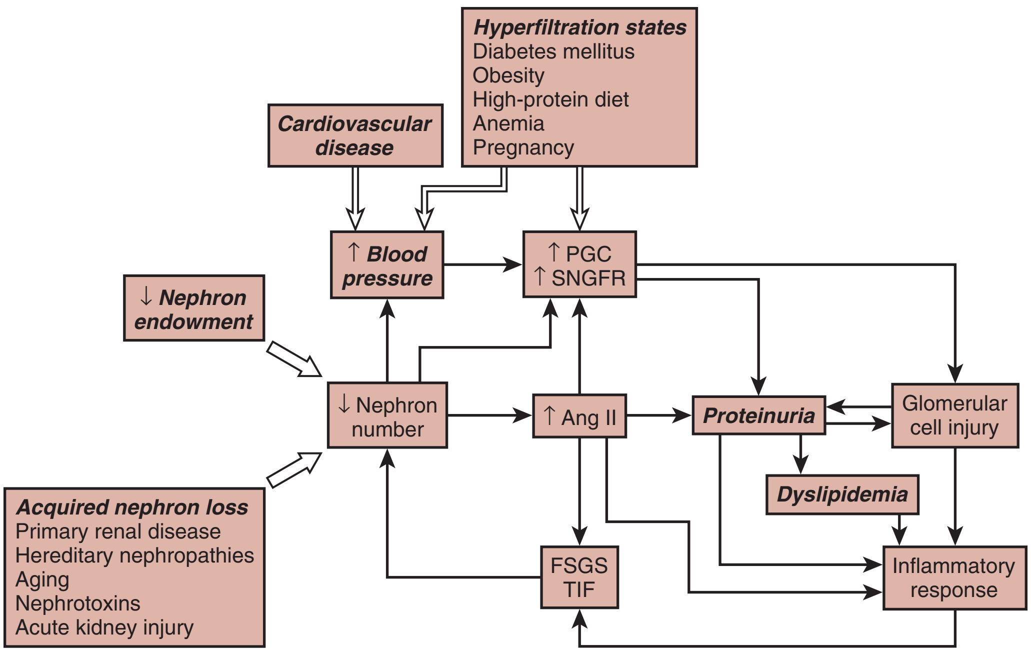

Fig. 20.1 Schematic representation showing the interaction of risk factors for chronic kidney disease progression with pathophysiologic mechanisms that contribute to a vicious cycle of progressive nephron loss. *Ang II,* Angiotensin II; *FSGS,* focal segmental glomerulosclerosis; *PGC,* glomerular capillary hydraulic pressure; *SNGFR,* single-nephron glomerular filtration rate; *TIF,* tubulointerstitial fibrosis. (Adapted from Taal MW, Brenner BM. Predicting initiation and progression of chronic kidney disease: developing renal risk scores. *Kidney Int.* 2006;70:1694–1705.)

Table 20.2 Risk Factors for Chronic Kidney Disease

Risk Factor	Susceptibility	Initiating	Progression
Older age	+		
Gender	+		
Ethnicity	+		+
Family history of chronic kidney disease	+		
Metabolic syndrome	+		
Hemodynamic factors			
Low nephron number	+		+
Diabetes mellitus	+	+	+
Hypertension	+		+
Obesity	+		+
High protein intake	+		+
Pregnancy		+	+
Primary renal disease		+	
Genetic renal disease		+	
Urologic disorders		+	
Acute kidney injury		+	+
Cardiovascular disease	+		+
Albuminuria			+
Hypoalbuminemia			+
Anemia	+		+
Dyslipidemia	+		+
Hyperuricemia	+		+
↑ Asymmetric dimethylarginine			+
Hyperphosphatemia			+
Low serum bicarbonate			+
Smoking	+		+
Nephrotoxins		+	+

example is hypertension, which exacerbates raised intraglomerular hydraulic pressure and therefore accelerates glomerular damage. Studies investigating progression factors should recruit subjects with relatively early stage CKD in a cohort study design. Nevertheless, distinguishing among these categories may in some cases be difficult because some factors [e.g., diabetes mellitus (DM)] may act in all three ways and, in some studies, it may be impossible to separate susceptibility factors from progression factors due to inadequate characterization of participants at study entry.

DEMOGRAPHIC VARIABLES

AGE

The prevalence of CKD increases with age and is reported to be as high as 56% in those 75 years or older.[18] Longitudinal studies of subjects without kidney disease have observed a decline in GFR with increasing age in some subjects, implying that nephron loss may be regarded as part of normal aging.[19] By contrast, aging is associated with an increase in several other risk factors for CKD, including hypertension, obesity, and CVD, which may contribute to the rise in CKD prevalence. Several population-based studies have found a higher incidence of proteinuria and CKD[20–22] as well as ESKD with increasing age.[23] Similarly, the incidence of a decline in renal function over 5 years was greater among older patients with hypertension.[24] One study reported that advanced age is a negative predictor of ESKD among patients with CKD, although older age was associated with a greater rate of decline in GFR.[25] This apparent contradiction is most likely explained by the competing risks of death and ESKD in older patients, illustrated by the observation from one longitudinal study that for patients 65 years and older, the risk of ESKD exceeded the risk of death only when the GFR was ≤15 mL/min/1.73 m^2 (Fig. 20.2).[26] By contrast, it has been found that a more rapid decrease of eGFR was associated with CKD in persons of a

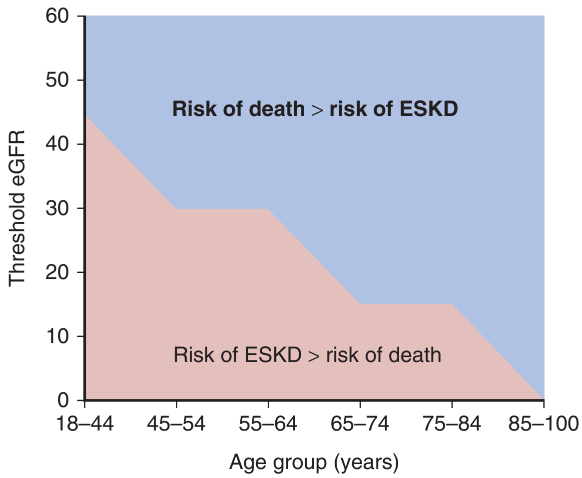

Fig. 20.2 Baseline estimated glomerular filtration rate (eGFR) threshold. Below this the risk for end-stage kidney disease (ESKD) exceeded the risk for death in each age group among 209,622 US veterans with chronic kidney disease stages 3 to 5 followed for a mean of 3.2 years. (From O'Hare AM, Choi AI, Bertenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol.* 2007;18:2758–2765.)

younger age, which in turn is associated with an increase in the 1-year mortality rate.[11]

A large individual-level meta-analysis that included data from 2,051,244 participants in 46 cohort studies has provided robust data regarding the effect of age on the risks associated with CKD. In the general population and high cardiovascular risk cohorts, the increase in relative risk of mortality associated with lower GFR declined with increasing age, but the increase in absolute risk of death was higher in older age groups. Similar trends were observed for the mortality risks associated with albuminuria. By contrast, the relative increase in risk of mortality did not decrease with increasing age in data from CKD cohorts. Furthermore, there was no attenuation of the risk of ESKD with increasing age in any of the cohort categories.[27]

Thus older age is a susceptibility factor for CKD, and the associated increase in risk of death and ESKD is observed at all ages. These observations suggest that targeted screening for CKD in older subjects would be a cost-effective strategy, but further studies are required to investigate the extent to which the risks associated with CKD in older adults may be attenuated by intervention. For further discussion of CKD in older age groups, see Chapters 22 and 84.

GENDER

Data regarding the effect of gender on the risk of CKD and progression are somewhat contradictory. Studies have reported a higher incidence of proteinuria and CKD among men in the general population and an increased risk of ESKD or death associated with CKD,[20,23,28] a higher risk of decline in renal function in male hypertensive patients,[24] a lower risk of ESKD in female patients with CKD stage 3,[25] and a shorter time to renal replacement therapy (RRT) in male patients with CKD stage 4 or 5.[29] In addition, most national registries, including the US Renal Data Service (USRDS), have reported a substantially higher incidence of ESKD in males [413 per million population (pmp) in 2003] versus females (280 pmp).[30] Previous meta-analyses have, however, yielded conflicting results, with one reporting a higher rate of decline in GFR in men[31] and another reporting a higher risk of doubling of the serum creatinine level or ESKD in women after adjustment for baseline variables, including blood pressure and urinary protein excretion.[32]

The largest metaanalysis to date included data from studies of 2,051,158 participants investigating the impact of gender on CKD-related outcomes. Whereas the risk of all-cause and cardiovascular mortality was higher in men than women, the relative risk of mortality increased with lower GFR and higher albuminuria in both, and the slope of increase in risk was steeper in women than men. Importantly, the relative risk of ESKD increased with lower GFR and higher albuminuria in both sexes and there was no evidence of a difference in the increase in risk between men and women (Fig. 20.3).[33] One limitation of many of the studies quoted is that menopausal status of the women was not documented. Nevertheless, it is clear from the most robust data published that CKD is associated with at least the same relative increase in risk of death and ESKD in women as in men. The reasons for the higher absolute incidence of RRT in men versus women require further investigation, and may be related to treatment preferences for RRT rather than biological differences in disease progression. For further discussion of the impact of gender on CKD, see Chapters 19, 21, and 51.

ETHNICITY

African Americans are overrepresented in the US dialysis population, suggesting that ethnicity is a strong risk factor for the progression of CKD to ESKD. Population-based studies have found a higher incidence of ESKD among African Americans that was attributable only in part to socioeconomic and other known risk factors.[7,23,34,35] Similarly, the risk of early renal function decline (increase in serum creatinine ≥ 0.4 mg/dL) was approximately threefold higher [odds ratio (OR), 3.15; 95% CI, 1.86 to 5.33] among black versus white diabetic adults, but 82% of this excess risk was attributable to socioeconomic and other known risk factors.[36] The risk of renal function decline over 5 years among hypertensive patients was greater in African Americans,[24] and African ancestry was independently associated with a greater rate of GFR decline in the Modification of Diet in Renal Disease (MDRD) study.[10] Interestingly, data from the Reasons for Geographic and Racial Differences in Stroke (REGARDS) Cohort Study have shown a lower prevalence of eGFR (50 to 59 mL/min/1.73 m^2) among African Americans but a higher prevalence of eGFR (10 to 19 mL/min/1.73 m^2) among white subjects[37] suggesting that African-American ethnicity acts as a progression factor but not as a susceptibility factor. A 2012 report from the USRDS showed a substantially higher incidence of ESKD in African Americans (3.3 times higher than whites), Hispanics (1.5 times higher than non-Hispanics), and Native Americans (1.5 times higher than whites).[38] Similarly, the prevalence of ESKD in 2012 was higher among minority groups: African Americans, 5671 pmp; Native Americans, 2600 pmp; Hispanics, 2932 pmp; Asians, 2272 pmp; and whites, 1432 pmp.[38] CKD and ESKD have also been reported to be more prevalent in other ethnic groups, including Asians,[39] Hispanics,[40] Native Americans,[41] Mexican Americans,[42] and Aboriginal Australians.[43] A large meta-analysis that included data from 940,366 participants in 25 general population cohort studies investigated the impact of ethnicity on the risks associated with CKD in blacks, whites, and Asians. The absolute risk of all-cause or cardiovascular mortality (after adjustment for age) and ESKD was higher in black versus white versus Asian participants. However, the relative risk of all-cause or cardiovascular mortality and ESKD increased to a similar degree with lower GFR or greater albuminuria in all the ethnic groups.[44]

Thus the risk between lower GFR or greater albuminuria and mortality or ESKD was not modified by ethnicity. The mechanisms underlying the associations between ethnicity and CKD remain to be elucidated, but possible explanations include genetic factors (see the "Hereditary Factors" section); increased prevalence of DM; lower nephron endowment; increased susceptibility to salt-sensitive hypertension; and environmental, lifestyle, and socioeconomic differences. Ethnicity and CKD are discussed further in Chapters 19, 21, and 51.

HEREDITARY FACTORS

Hereditary renal diseases resulting from a single gene defect, such as autosomal dominant polycystic kidney disease (PKD),

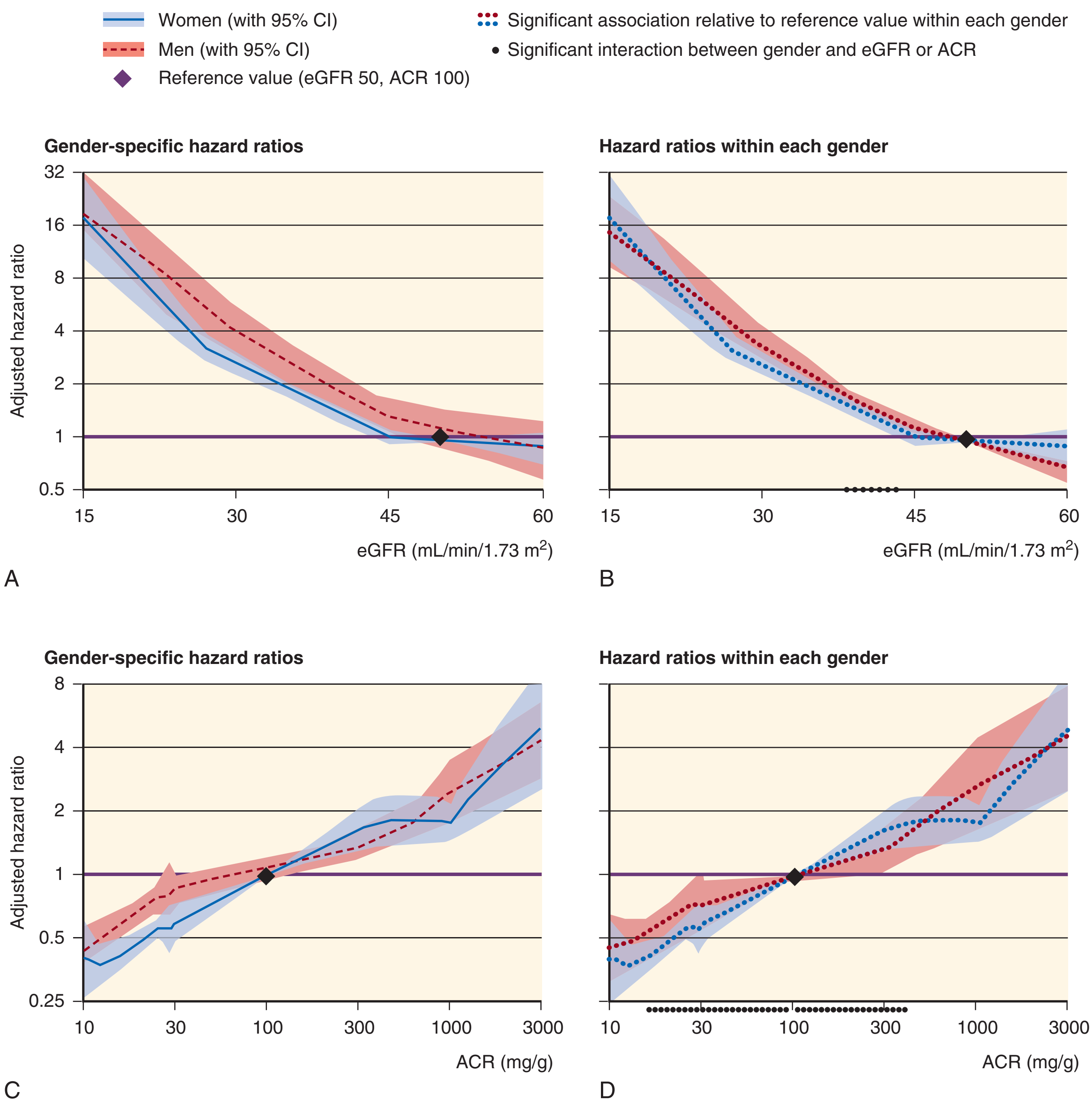

Fig. 20.3 (A and B) Hazard ratios of end-stage kidney disease according to estimated glomerular filtration rate (eGFR). (C and D) Urinary albumin-to-creatinine ratio (ACR) in men versus women in chronic kidney disease cohorts. (A and C) Gender-specific hazard ratios, including a main effect for male gender at the reference point. (B and D) Hazard ratios in each gender, thus visually removing the baseline difference between men and women. Hazard ratios were adjusted for age, gender, race, smoking status, systolic blood pressure, history of cardiovascular disease, diabetes, serum total cholesterol concentration, body mass index, and estimated glomerular filtration rate splines or albuminuria. *CI,* Confidence interval. (Adapted from Nitsch D, Grams M, Sang Y, et al. Associations of estimated glomerular filtration rate and albuminuria with mortality and renal failure by sex: a meta-analysis. *BMJ* 2013;346:f324.)

Alport disease, Fabry disease, and congenital nephrotic syndrome, account for a relatively small yet clinically important proportion of all patients with CKD. Nevertheless, evidence is rapidly accumulating that genetic factors account for familial clustering of many other forms of CKD with multifactorial causes. Among 25,883 incident ESKD patients, 22.8% reported a family history of ESKD,[45] and screening of the relatives of patients with ESKD revealed evidence of CKD in 49.3%.[46] In another case–control study, including 689 patients with ESKD and 361 controls, having one first-degree relative with CKD increased the risk of ESKD by 30% [hazards ratio (HR), 1.3; 95% CI, 0.7 to 2.6] and having two such relatives increased it tenfold (HR, 10.4; 95% CI, 2.7 to 40.2) after controlling for multiple known risk factors, including diabetes and hypertension.[47] Similarly, a case–control study of 103 US white patients with ESKD reported a 3.5-fold increase in risk of ESKD (95% CI, 1.5 to 8.4) with the presence of a first-, second-, or third-degree relative with ESKD.[48] A genetic explanation for the high incidence of ESKD observed in African Americans was provided by groundbreaking research that identified strong associations between ESKD and two coding variants in the apolipoprotein L1 (*APOL1*) gene. These gene variants confer resistance to infection with *Trypanosoma brucei rhodesiense*, which causes sleeping sickness. This observation explains how selection likely resulted in a high prevalence of these variants in the population. Subsequent studies have identified associations between *APOL1* risk variants and several renal pathologies, including focal segmental glomerulosclerosis (FSGS), human immunodeficiency virus (HIV)-associated nephropathy (HIVAN), sickle cell kidney disease, and severe lupus nephritis.[49] Moreover, cohort studies have reported associations between *APOL1* risk variants and risk of progression to ESKD. Risk of progression was the lowest in European Americans (with no risk variants), intermediate in African Americans (with no or one risk variant), and highest in African Americans (with two risk variants).[50] It is estimated that *APOL1* variants account for 40% of disease burden due to CKD in African Americans.

Despite the strong association between inheritance of two *APOL1* risk variants and ESKD, only a minority of people with this genotype actually develop kidney disease, suggesting that the action of a second factor is required to cause disease in genetically susceptible individuals. HIV is one example of such a second hit, but it has been proposed that other viruses and other gene variants may also be important.[49]

Other studies have suggested that genetic factors also increase susceptibility to early manifestations of CKD. In a study of 169 families with one type 2 diabetic proband, the diabetic siblings of probands with microalbuminuria had a significantly increased risk of also having microalbuminuria, after adjustment for confounding risk factors (OR, 3.94; 95% CI, 1.93 to 9.01), than the diabetic siblings of probands without microalbuminuria.[51] Furthermore, the nondiabetic siblings of diabetic probands with microalbuminuria had a significantly higher urinary albumin excretion rate (within the normal range) than the nondiabetic siblings of normoalbuminuric diabetic probands.

Genome-wide association studies (GWASs) have identified multiple novel loci that are significantly associated with serum creatinine levels or CKD.[52–54] Furthermore, a recent GWAS meta-analysis conducted in 63,558 participants of European descent identified significant associations between GFR decline over time and three gene loci – uromodulin (*UMOD*) (previously associated with CKD and ESKD), *GALNTL5/GALNT11,* and *CDH23*. It was estimated that the heritability of GFR decline in this population was 38%.[55] Further studies have investigated the role of epigenetic factors (heritable changes in the pattern of gene expression not attributable to changes in the primary nucleotide sequence) that may affect the risk of CKD progression. One study compared the genome-wide DNA methylation profile in 20 people from the Chronic Renal Insufficiency Cohort (CRIC) study with the most rapid decline in GFR and 20 with the most stable GFR. Results identified differences in the methylation of several genes associated with epithelial-to-mesenchymal transition and inflammation that may be involved in the mechanisms of CKD progression.[56]

From this discussion, it is clear that genetic factors may act as susceptibility factors in some subjects, initiating factors in those with CKD due to a single gene defect, or progression factors in others. The rapid growth in knowledge of genetic aspects of CKD will likely result in genetic risk factors becoming increasingly important in risk prediction for patients with CKD. For a more detailed discussion of genetic aspects of kidney disease, see Chapters 43–45.

HEMODYNAMIC FACTORS

Experimental studies have shown that glomerular hemodynamic responses (e.g., glomerular capillary hypertension and hyperfiltration) to nephron loss[57] and chronic hyperglycemia[58] are critical factors in establishing the vicious cycle of nephron loss characteristic of CKD. In addition, any factor that further increases glomerular hypertension and/or hyperfiltration may be expected to exacerbate glomerular damage and accelerate the progression of CKD (see Fig. 20.1).

DECREASED NEPHRON NUMBER

NEPHRON ENDOWMENT

Autopsy studies have revealed that the number of nephrons per kidney varies widely in humans, from 210,332 to 2,702,079 in one series.[59] Multiple factors have been shown to influence nephron endowment, including those that affect the feto-maternal environment as well as genetic factors.[60] A substantial body of evidence supports the hypothesis that low nephron endowment predisposes individuals to CKD by provoking an increase in SNGFR and, therefore, a reduction in renal reserve. The ascertainment of nephron number in living human subjects is currently not possible, but autopsy studies have shown an association between reduced nephron number and hypertension,[61] as well as glomerulosclerosis.[62] In human autopsy studies, low birth weight is directly associated with reduced nephron number,[63,64] and birth weight may therefore serve as a marker of nephron endowment. Low birth weight is also a risk factor for later-life hypertension and DM, both of which further increase the risk of CKD.[65] One meta-analysis of 32 studies, which included data from >2 million subjects, reported a significantly increased risk of albuminuria (OR, 1.81; 95% CI, 1.19 to 2.77) and ESKD (OR, 1.58; 95% CI, 1.33 to 1.88) associated with low birth weight.[66] Thus low birth weight, acting as a marker of reduced nephron endowment, may be regarded as a susceptibility and progression risk factor for CKD. Factors affecting nephron endowment and the consequences of reduced nephron endowment are discussed in more detail in Chapter 21.

ACQUIRED NEPHRON DEFICIT

In experimental models of acquired nephron deficit, severe nephron loss (5/6 nephrectomy) alone initiates a cycle of progressive injury in the remaining glomeruli, mediated primarily through glomerular hypertension and hyperfiltration.[57] In 14 patients subjected to similarly large reductions in nephron number following partial resection of a single kidney, two developed ESKD and nine developed proteinuria, the extent of which was inversely correlated with the amount of renal tissue remaining.[67] Lesser degrees of acquired nephron loss, such as removal of one of two previously normal kidneys (uninephrectomy), may not be sufficient to cause CKD in most subjects.[15,68,69] However, nephrectomy for renal cell carcinoma is associated with an increased risk of developing CKD that is greater after radical nephrectomy than partial nephrectomy, suggesting that in the presence of subclinical kidney damage, acquired nephron loss may provoke CKD, and that the risk is proportional to the number of nephrons removed.[70]

Nephron loss may also predispose individuals to CKD if they are also exposed to other risk factors. This is perhaps best illustrated by the observation that uninephrectomy exacerbates renal injury in experimental diabetic nephropathy[71] and, in persons with diabetes, uninephrectomy increases the risk of developing diabetic nephropathy.[17]

The interaction between nephron loss and other risk factors is further illustrated by the observation that in a study of 488 people who had surgery for renal cell carcinoma, radical nephrectomy (compared with partial nephrectomy), diabetes, and increased age were each independently associated with an increased risk of developing CKD at least 6 months after surgery. In those who had a partial nephrectomy but no additional risk factors, only 7% developed CKD, but this increased to 24%, 30%, and 42% in those 60 years or older, those with hypertension, and diabetes, respectively.[72]

In most forms of CKD, initial nephron loss due to primary renal disease, multisystem disorders that involve the kidney, or exposure to nephrotoxins is focal, but hemodynamic adaptations in the remaining glomeruli are thought to contribute to nephron loss by provoking further glomerulosclerosis (see Chapter 51). Several epidemiologic studies have supported this hypothesis by showing that patients with a reduced GFR are at increased risk of a further decline in renal function. Two large meta-analyses of cohort studies identified baseline GFR as a strong predictor of ESKD. Among 845,125 participants from the general population, the eGFR was independently associated with an increased risk of developing ESKD when it fell below 75 mL/min/1.73 m^2. For groups of patients with average eGFRs of 60, 45, and 15 mL/min/1.73 m^2, the hazard ratios for developing ESKD were 4, 29, and 454, respectively, when compared with a reference group with an eGFR of 95 mL/min/1.73 m^2. Similar findings were reported in a further 173,892 participants selected for being at increased risk of developing CKD (Fig. 20.4).[73] Among 21,688 patients selected for having CKD, a lower eGFR was an independent risk factor for ESKD, such that a fall of 15 mL/min/1.73 m^2 below a threshold of 45 mL/min/1.73 m^2 was associated with a pooled hazard ratio of 6.24.[74] Further analyses by the CKD Prognosis Consortium have confirmed that the association between reduced GFR and increased risk of ESKD persists independently of gender,[33] age,[27] ethnicity,[44] diabetes,[75] and hypertension.[76] In addition, analysis of data from 1,530,648 participants has shown that change in GFR over time is strongly predictive of future risk

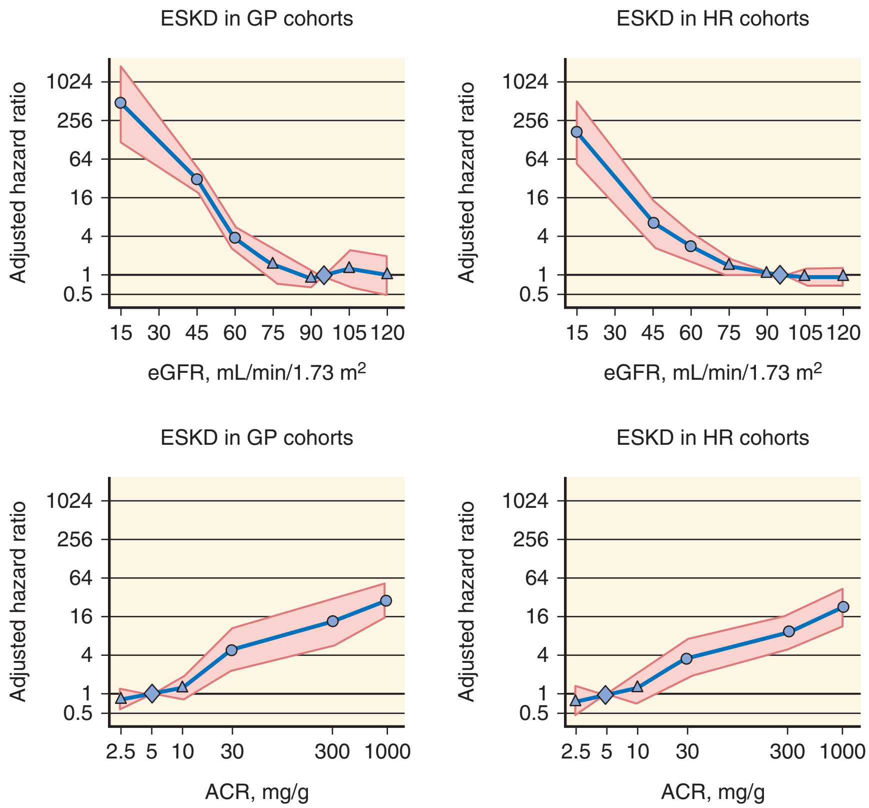

Fig. 20.4 Pooled hazard ratios (95% confidence interval) for end-stage kidney disease (ESKD) according to spline estimated glomerular filtration rate (eGFR) *(upper panels)* and albumin-to-creatinine ratio *(lower panels),* adjusted for each other and for age, gender, and cardiovascular risk factors (continuous analyses). Reference categories are eGFR of 95 mL/min/1.73 m^2 and albumin-to-creatinine ratio of 5 mg/g or dipstick negative or trace. *(Left panels)* Results for general population cohorts. *(Right panels)* High-risk cohorts. Dots represent statistical significance, triangles represent nonsignificance, and shaded areas are 95% confidence interval. *ACR,* Albumin-to-creatinine ratio; *GP,* general population; *HR,* high risk. (From Gansevoort RT, Matsushita K, van der Velde M, et al. Chronic Kidney Disease Prognosis Consortium: lower estimated GFR and higher albuminuria are associated with adverse kidney outcomes in both general and high-risk populations. A collaborative meta-analysis of general and high-risk population cohorts. *Kidney Int.* 2011;79:1341–1352.)

of ESKD (and mortality), suggesting that a GFR decline of 30% may be useful as a surrogate marker of CKD progression in clinical trials.[77] Thus, in different contexts, acquired nephron deficit may be regarded as a susceptibility factor (e.g., after donor nephrectomy in a healthy kidney donor), initiation factor (when severe nephron loss provokes glomerulosclerosis in remaining previously normal glomeruli), or progression factor (when nephron loss accelerates preexisting damage in remaining glomeruli).

The importance of GFR as a risk factor has emphasized the need for more accurate methods to estimate it. Adoption of the MDRD equation improved detection of CKD and made possible much of the epidemiologic research on CKD, but it was recognized from the outset that the MDRD equation was imperfect and, in particular, tended to underestimate true GFR at values >60 mL/min/1.73 m^2. This is important because this is the threshold below which CKD may be diagnosed without other evidence of kidney damage. Several other equations have been developed to estimate GFR from serum creatinine concentration, culminating in a recommendation by KDIGO that the MDRD equation should be replaced by the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation, which is more accurate than the MDRD equation and results in less bias, particularly at GFR values >60 mL/min/1.73 m^2. Further analysis by the CKD Prognosis Consortium found that the eGFR determined by the CKD-EPI equation results in a lower prevalence of CKD stages 3 to 5 (8.7% vs. 6.3%) and affords better risk prediction than the MDRD equation. Among those classified as having an eGFR of 59 to 45 mL/min/1.73 m^2 by the MDRD equation, 34.7% were reclassified by the CKD-EPI as having an eGFR of 89 to 60 mL/min/1.73 m^2, and those reclassified had a lower incidence of adverse outcomes versus those not reclassified (e.g., incidence of all-cause mortality 9.9 vs. 34.5/1000 person-years).[78]

The limitations of creatinine as a marker of GFR due to nonrenal factors that affect serum creatinine concentration, including muscle mass and diet, have prompted a search for alternatives. Cystatin C, a peptide produced by all nucleated cells and therefore not affected by muscle mass, has emerged as the most promising alternative. The production of reference material to standardize cystatin C assays has greatly improved the potential for clinical application, and equations have been developed to estimate the GFR from the serum cystatin C concentration or from creatinine and cystatin C levels together. The combined equation and CKD-EPI (creatinine) equation have similar bias, but the combined equation yields better precision and accuracy.[79] Further work by the CKD Prognosis Consortium has reported that when cystatin C is used to estimate GFR, reclassification to a higher GFR category (higher than that assigned by eGFR creatinine) is associated with a lower risk of all-cause mortality, cardiovascular mortality, and ESKD.[80] It should be noted, however, that in this analysis, CKD was defined by a single eGFR value. In one analysis that included 1741 persons with CKD recruited from primary care on the basis of two abnormal eGFR results, the use of cystatin C resulted in reclassification of only 7.7% as not having CKD and a large proportion (59%) were reclassified as having a more advanced stage of CKD with no improvement in risk prediction.[81] Further studies are warranted to further evaluate the potential benefits of using cystatin C for estimating GFR.

ACUTE KIDNEY INJURY

Despite previous perceptions that patients who recover from acute kidney injury (AKI) regain normal renal function and have a good prognosis, several cohort studies have reported that recovery from AKI is associated with a substantially increased risk of CKD and death. Among 3769 adults who required dialysis for AKI and survived dialysis free for at least 30 days, the incidence rate for chronic dialysis was 2.63/100 versus 0.91/100 person-years in 13,598 matched controls (adjusted HR, 3.23; 95% CI, 2.70 to 3.86).[82] The relative risk was particularly high for those with no previous diagnosis of CKD (adjusted HR, 15.54; 95% CI, 9.65 to 25.03). There was no difference in survival between the groups. In another study of similar design, outcomes were investigated in 343 patients with a preadmission eGFR > 45 mL/min/1.73 m^2 who required dialysis for AKI but survived for at least 30 days after discharge without dialysis. After controlling for potential confounders, AKI that required dialysis was associated with a 28-fold increase in the risk of developing CKD stage 4 or 5 (adjusted HR, 28.1; 95% CI, 21.1 to 37.6) and more than double the risk of death (adjusted HR, 2.3; 95% CI, 1.8 to 3.0) versus 555,660 adult patients hospitalized during the same period but without AKI.[83]

Analysis of data from a cohort of 233,803 Medicare beneficiaries 67 years or older who were hospitalized in 2000 reported a substantially increased risk of developing ESKD in those who developed AKI on a background of CKD (HR, 41.2; 95% CI, 34.6 to 49.1) or without previous CKD (HR, 13.0; 95% CI, 10.6 to 16.0) versus those who did not develop AKI. The importance of AKI as a risk factor for CKD initiation was further illustrated by the observation that among patients who had AKI without preexisting CKD ($N = 4730$), 72.1% developed CKD within 2 years of the AKI episode. Furthermore, 25.2% of those who developed ESKD had a history of AKI.[84] In a similar study, a cohort of 113,272 patients hospitalized with a primary diagnosis of acute tubular necrosis (ATN), AKI, pneumonia, or myocardial infarction (control group) was studied. Overall, 11.4% progressed to CKD stage 4 during follow-up, including 20.0% of those with ATN, 13.2% of those with AKI, 24.7% of those with preexisting CKD, and 3.3% of the control patients. After controlling for other variables, having a diagnosis of AKI, ATN, or CKD increased the risk of developing CKD stage 4 by 303%, 564%, and 550%, respectively, versus controls. After controlling for covariates, AKI and CKD were associated with an increased risk of death of 12% and 20%, respectively, versus controls.[85]

The multiplicative effect of AKI on CKD progression is further illustrated by a study of 39,805 patients with an eGFR < 45 mL/min/1.73 m^2 prior to hospitalization. Those who survived an episode of dialysis-requiring AKI had a very high risk of developing ESKD within 30 days of hospital discharge (i.e., nonrecovery of AKI) that was related to the preadmission eGFR. For an eGFR of 30 to 44 mL/min/1.73 m^2, the incidence of ESKD was 42%, and for an eGFR of 15 to 29 mL/min/1.73 m^2, it was as high as 63%, whereas the incidence of ESKD was only 1.5% in those who did not have dialysis-requiring AKI. In patients who survived longer than 30 days after hospital discharge without ESKD, the incidence of ESKD and death at 6 months was 12.7% and 19.7%, respectively, versus 1.7% and 7.4% in the comparator group with CKD but no AKI. After adjustment for multiple risk factors, AKI was associated with a 30% increase in long-term risk for death or ESKD (adjusted HR, 1.30; 95% CI, 1.04 to 1.64).[86]

Consistent with the findings of individual studies, a meta-analysis of 13 cohort studies reported a significantly increased risk of developing CKD and ESKD in patients who had survived an episode of AKI versus participants without AKI (pooled adjusted HR for CKD, 8.8; 95% CI, 3.1 to 25.5; pooled HR for ESKD, 3.1; 95% CI, 1.9 to 5.0).[87] Taken together, these data show that AKI should be regarded as an important risk factor for CKD initiation and progression. The mechanisms responsible for these observations require further elucidation, but have been proposed to include nephron loss, loss of peritubular capillaries, cell cycle arrest, cell senescence, pericyte and myofibroblast activation, fibrogenic cytokine production, and interstitial fibrosis.[88,89]

The incidence of AKI has increased, and it is likely to become an increasingly important risk factor for CKD among older patients.

HYPERTENSION

Hypertension, which is defined as an increased systolic blood pressure >140 mm Hg and/or diastolic blood pressure >90 mm Hg, or the need for pharmacologic therapy to achieve BP targets, is an almost universal consequence of reduced renal function.[90] From a CKD-focused standpoint, this is due to sodium retention, hypervolemia, and neurohormonal activation associated with progressive decrease in GFR.[91–93] However, hypertension itself is also known to be an important factor in the progression of CKD toward ESKD. In the hypothesis of CKD progression presented in Fig. 20.1, it is clear that elevated systemic blood pressure transmitted to the glomerulus would contribute to glomerular hypertension and thus accelerate glomerular damage. Hypertension has been shown to be predictive of ESKD risk in several large population-based studies.[20,21,28,94] Furthermore, a close association between the magnitude of increased risk and level of blood pressure has been reported in several studies, so that even elevations in blood pressure below the threshold for the diagnosis of hypertension were associated with increased risk of ESKD.[20,28,95]

Among patients with CKD in the MDRD study, higher baseline mean arterial pressure (MAP) independently predicted a greater rate of GFR decline.[10] These observations have led to the suggestion that blood pressure be viewed as a continuous rather than dichotomous risk factor for CKD, with less emphasis on traditional definitions of hypertension and normotension.[96] Whereas the primary analysis of data from the MDRD study found no significant difference between the rate of decline in GFR between patients randomized to intensive blood pressure control (target MAP < 92 mm Hg, equivalent to <125/75 mm Hg) and standard blood pressure control (target MAP < 107 mm Hg, equivalent to 140/90 mm Hg), a secondary analysis did show benefit associated with the low blood pressure target in patients with higher levels of baseline proteinuria.[97] Further secondary analysis showed that the lower achieved blood pressure was also associated with a slower GFR decline, an effect that was more marked in patients with higher baseline proteinuria.[98] Thus the MDRD study results suggest a significant interaction between blood pressure and proteinuria as risk factors for CKD progression.

The Systolic Blood Pressure Intervention Trial, which randomized >9000 subjects, including about one-third with nondiabetic CKD, was able to determine that targeting systolic blood pressure of <120 mm Hg, as compared with <140 mm Hg, in high-risk CVD patients without diabetes resulted in fewer fatal and nonfatal CVEs and death.[99] A subgroup analysis of the 2646 participants with CKD showed no difference in the primary renal outcome of ≥50% decline in GFR or ESKD, though there was a greater decrease in GFR after 6 months (–0.47 versus –0.32 mL/min/1.73 m^2 per year) associated with the lower BP target. The risks of all-cause mortality (HR, 0.72; 95% CI, 0.53 to 0.99) and CVE (HR, 0.81; 95% CI, 0.63 to 1.05) were lower in the low BP target group and there was no overall increase in serious adverse events, though AKI, hypokalemia, and hyperkalemia were increased in the low BP target group.[100] In the HALT Progression of Polycystic Kidney Disease (HALT-PKD) study, targeting of a blood pressure of ≤120/70 mm Hg in patients with autosomal dominant polycystic kidney disease (ADPKD) through use of lisinopril and telmisartan allowed for a slower increase in total kidney volume, no overall change in eGFR, a greater decline in left ventricular mass index, and a greater reduction in albuminuria.[101]

Taken together, there is convincing evidence that elevated blood pressure is an important risk factor for progression of CKD, although unequivocal evidence from randomized controlled trials is lacking, and some uncertainty remains regarding optimal treatment targets.

OBESITY AND METABOLIC SYNDROME

In experimental models, obesity is associated with hypertension, proteinuria, and progressive renal disease. Micropuncture studies have confirmed that obesity is another cause of glomerular hyperfiltration and glomerular hypertension that can be predicted to exacerbate the progression of CKD.[102,103] Furthermore, several other factors associated with obesity and the metabolic syndrome may contribute to renal damage, including hormones and proinflammatory molecules produced by adipocytes,[104] increased mineralocorticoid levels and/or mineralocorticoid receptor activation by cortisol,[105] and reduced adiponectin levels.[106] In humans, severe obesity is associated with increased renal plasma flow, glomerular hyperfiltration, and albuminuria, abnormalities that are reversed by weight loss.[107] Obesity, as defined by increased body mass index (BMI), has been associated with increased risk of developing CKD in several large population-based studies.[21,108,109] Furthermore, one study has found a progressive increase in relative risk of developing ESKD associated with increasing BMI (relative risk, 3.57; 95% CI, 3.05 to 4.18 for BMI of 30.0 to 34.9 kg/m^2 vs. BMI of 18.5 to 24.9 kg/m^3) among 320,252 subjects confirmed to have no evidence of CKD at initial screening.[110]

There is evidence that obesity may directly cause a specific form of glomerulopathy characterized by proteinuria and histologic features of focal and segmental glomerulosclerosis,[111,112] but it is likely that it also acts as a risk factor in the development of several other forms of renal disease. One study has identified childhood obesity as a risk factor for CKD in adulthood. Among 4340 participants born in one week in 1946, pubertal-onset obesity and obesity throughout childhood were associated with an increased risk of CKD (defined by eGFR < 60 mL/min/1.73 m^2 or albuminuria) at age 60 to 64 years.[113] Interest has also focused on the role of the metabolic syndrome (insulin resistance), as defined by the presence of abdominal obesity, dyslipidemia, hypertension, and fasting hyperglycemia, in the development of CKD. An analysis of data from the Third National Health and Nutrition Examination Survey (NHANES III) data found a significantly increased risk of CKD and microalbuminuria in subjects with the metabolic syndrome as well as a progressive increase in risk associated with the number of components of the metabolic syndrome present.[114] Furthermore, a large longitudinal study of 10,096 patients without diabetes or CKD at baseline identified metabolic syndrome as an independent risk factor for the development of CKD over 9 years (adjusted OR, 1.43; 95% CI, 1.18 to 1.73). Again, there was a progressive increase in risk associated with the number of traits of the metabolic syndrome present (OR, 1.13; 95% CI, 0.89 to 1.45 for one trait vs. OR, 2.45; 95% CI, 1.32 to 4.54 for five traits).[115] In another study, patient hip-to-waist ratio, a marker of insulin resistance, was independently associated with impaired renal function, even in lean individuals (BMI < 25 kg/m^3), among a population-based cohort of 7676 subjects.[116]

It is widely recognized that weight loss is difficult to achieve in obese individuals, but surgical intervention in the form of gastric banding or bypass appears to offer the most effective long-term outcomes. Beneficial renoprotective effects of weight loss have been reported in a meta-analysis of observational studies that found an association between weight loss and reduction in proteinuria independent of blood pressure,[117] as well as smaller studies that reported improvement or stabilization of renal function[118] or reduction in proteinuria[119] following bariatric surgery in subjects with CKD.

The best method for assessing obesity in CKD remains to be determined. A further systematic review analyzed the effects of weight loss achieved by bariatric surgery, medication, or diet in 31 studies and found that in most studies, weight loss was associated with reductions in proteinuria. In people with glomerular hyperfiltration, the GFR tended to decrease with weight loss, and in those with a reduced GFR, it tended to increase.[120] BMI is the most widely applied method but does not take body composition into account. One study has reported a high sensitivity but relatively low specificity of BMI to detect obesity in subjects with CKD.[121]

HIGH DIETARY PROTEIN INTAKE

Consistent with the hypothesis that the glomerular hemodynamic changes associated with hyperfiltration accelerate glomerular injury, experimental studies have reported that a high-protein diet accelerates renal disease progression, whereas dietary protein restriction[122,123] results in normalization of glomerular capillary hydraulic pressure as well as SNGFR and marked attenuation of glomerular damage.[57] Observational studies in humans have reported an increased risk of microalbuminuria associated with higher dietary protein intake in subjects with diabetes and hypertension (OR, 3.3; 95% CI, 1.4 to 7.8), but not in healthy subjects or those with isolated diabetes or hypertension,[124] again illustrating the interaction between risk factors for CKD. In another study, high intake of protein, particularly nondairy animal protein, was associated with a greater rate of GFR decline among women with an eGFR of 80 to 55 mL/min/1.73 m^2 but not in those with an eGFR of >80 mL/min/1.73 m^2.[125] Randomized trials investigating the effects of high-protein diet are lacking, but several studies have sought to examine the potential renoprotective effects of dietary protein restriction. In the MDRD study, primary analysis revealed no significant difference in the mean rate of GFR decline in subjects randomized to low- or very-low-protein diets,[97] but secondary analysis of outcomes according to achieved dietary protein intake indicated that a reduction in protein intake of 0.2 g/kg/day correlated with a 1.15-mL/min/year reduction in the rate of GFR decline, equivalent to a 29% reduction in the mean rate of GFR decline.[126] By contrast, long-term follow-up of participants in study 2 of the MDRD trial found no renoprotective benefit among those randomized to very-low-protein diet in the original study, but did report a higher risk of death in this group (HR, 1.92; 95% CI, 1.15 to 3.20).[127] As such, dietary protein restriction cannot be routinely recommended for patients with CKD. Additional discussion on the role of dietary protein restriction in the management of CKD is available in Chapters 51 and 60.

PREGNANCY AND PREECLAMPSIA

Physiologic adaptations during pregnancy provoke glomerular hyperfiltration that usually does not cause renal damage. In the context of preexisting CKD, however, the glomerular hyperfiltration of pregnancy can be predicted to exacerbate proteinuria and glomerular injury. Several studies have shown an increased risk of CKD progression during pregnancy, particularly when the pregestational serum creatinine is ≥1.4 mg/dL (≥124 μmol/L). In one study of 82 pregnancies in 67 women with primary renal disease and serum creatinine level of ≥1.4 mg/dL, blood pressure, serum creatinine, and proteinuria increased during pregnancy. In 70 pregnancies with postpartum data available, persistent loss of maternal renal function at 6 months was reported in 31%, and by 12 months eight women had progressed to ESKD. Adverse obstetric outcomes included preterm delivery in 59% and low birth weight in 37%, although fetal survival was 93%.[128]

In a more recent series of 49 women with CKD stage 3 to 5 before pregnancy, the mean GFR declined during pregnancy (from 35 ± 12.2 to 30 ± 13.8 mL/min/1.73 m^2), but there was no change in the mean postpartum rate of GFR decline. Nevertheless, a pregestational GFR < 40 mL/min/1.73 m^2, combined with proteinuria of >1 g/day, was associated with a more rapid postpartum GFR decline and a shorter time to ESKD or halving of GFR and low birth weight.[129] Although earlier reports suggested good outcomes, one recent study has reported adverse effects associated even with early stage CKD. In 91 pregnancies with predominantly CKD stages 1 and 2, modest increases in hypertension, serum creatinine, and proteinuria were observed.[130] An increase in adverse obstetric outcomes, including preterm delivery, lower birth weight, and admission to a neonatal intensive care unit versus low-risk pregnancy controls was also reported; this remained true, even when only those with CKD stage 1 were considered, although there were no perinatal deaths. By contrast, pregnancy was not associated with a more rapid decline in the GFR over 5 years in a cohort of 245 women of childbearing age with IgA nephropathy and serum creatinine level of ≤1.2 mg/dL (in the majority).[131]

Complications of pregnancy and, in particular, hypertension and preeclampsia, may also cause renal damage. In one large population-based study, renal outcomes were assessed in 570,433 women who had had at least one singleton pregnancy. Only 477 women developed ESKD at a mean of 17 ± 9 years after the first pregnancy (overall rate, 3.7/100,000 women/year), but preeclampsia was associated with a significant increase in the risk of ESKD, ranging from a relative risk of 4.7 for preeclampsia in a single pregnancy (95% CI, 3.6 to 6.1) to a relative risk of 15.5 for preeclampsia in two or three pregnancies (95% CI, 7.8 to 30.8). The risk was further increased if the pregnancy resulted in a low-birth-weight or preterm infant. Causes of ESKD were glomerulonephritis in 35%, hereditary or congenital disease in 21%, diabetic nephropathy in 14%, and interstitial nephritis in 12%.[132] Similarly, in women with diabetes prior to pregnancy, preeclampsia and preterm birth were associated with significantly increased risks of ESKD and death, illustrating how different risk factors for CKD may interact to increase risk.[133]

A large cohort study reported an increased risk of multiple adverse health outcomes after hypertension during pregnancy, including CVD, DM, and CKD (HR, 1.91; 95% CI, 1.18 to 3.09).[134] Similarly, a very large case–control study found that hypertension during pregnancy was associated with a substantially increased risk of subsequent CKD (HR, 9.38; 95% CI, 7.09 to 12.4) or ESKD (HR, 12.4; 95% CI, 8.53 to 18.0).[135] In both these studies, the risks of CKD were substantially higher if preeclampsia developed during the pregnancy. Possible explanations for these observations include the presence of pathogenic factors common to CKD and preeclampsia, including obesity, hypertension, insulin resistance, and endothelial dysfunction; exacerbation by preeclampsia of preexisting subclinical CKD; and effects of preeclampsia on the kidney that increase the risk of CKD later in life.[136] That preeclampsia may provoke renal damage has been suggested by several studies showing an increased incidence of microalbuminuria after preeclampsia. A meta-analysis of seven of these studies reported a 31% prevalence of microalbuminuria at a weighted mean of 7.1 years after preeclampsia versus 7% in a control group with uncomplicated pregnancies.[137] Further research is required to identify which mechanisms are most relevant but, even without further information, preeclampsia should be regarded as a risk factor for the development and progression of CKD.

MULTISYSTEM DISORDERS

DIABETES MELLITUS

Diabetic nephropathy has rapidly become the single most common cause of ESKD worldwide. Diabetes was associated with a substantially increased risk of ESKD or death associated with CKD in one population-based study of 23,534 subjects (HR, 7.5; 95% CI, 4.8 to 11.7),[28] as well as an increased risk of moderate CKD (estimated creatinine clearance < 50 mL/min) in another study of 1428 subjects with an estimated creatinine clearance of >70 mL/min at baseline.[138] Evidence that glycemic control is a key risk factor for the development of diabetic nephropathy has been shown in randomized trials that found a reduced risk of developing nephropathy in subjects with type 1[139] and type 2[140] diabetes randomized to tight glycemic control. The pathogenesis of diabetic nephropathy is complex and involves multiple mechanisms, including glomerular hemodynamic factors,[58,141] advanced glycation end product formation, generation of reactive oxygen species, and upregulation of profibrotic growth factors and cytokines.[142,143] In at least one study, diabetic nephropathy was associated with more rapid progression to ESKD than other causes of CKD.[29,144,145] Thus diabetes may be regarded as a susceptibility, initiation, and progression risk factor for CKD. For further discussion of the pathogenesis of diabetic nephropathy, see Chapter 39.

PRIMARY RENAL DISEASE

Whereas substantial variation in the rate of GFR decline has been observed among subjects with a common cause of CKD, there is also evidence that some forms of CKD may provoke more rapid progression than others. In the MDRD study[10] and the Chronic Renal Insufficiency Standards Implementation Study (CRISIS),[144] a diagnosis of ADPKD was an independent predictor of a greater rate of GFR decline. In several cohort studies, diabetic nephropathy was associated with shorter time to ESKD[29] or a more rapid GFR decline than other diagnoses.[144,145]

CARDIORENAL SYNDROME

Multiple studies have reported that CKD is associated with a substantial increase in the risk of CVD,[146] and it is therefore not surprising that CVD is also associated with an increased risk of CKD. Among hospitalized Medicare beneficiaries, the prevalence of CKD stage 3 or worse was 60.4% for those with heart failure and 51.7% for those with myocardial infarction. The presence of CKD in addition to heart disease was associated with a significantly increased risk of death and progression to ESKD.[147] These observations may in part be explained by the fact that CVD and CKD share many risk factors, including obesity, metabolic syndrome, hypertension, DM, dyslipidemia, and smoking. In addition, CVD may exert effects on the kidneys that promote the initiation and progression of CKD, including decreased renal perfusion in heart failure and atherosclerosis of the renal arteries. This is a phenomenon known as cardiorenal syndrome (CRS), which is an acute or chronic dysfunction in one organ system leading to dysfunction in the other, specifically in the cardiovascular and renal systems. Mechanisms that lead to the development of CRS include neurohormonal activation, altered renal blood flow, renal congestion, and right ventricular dysfunction.[148] A typical example of CRS is seen in renal atherosclerosis, which was detected in 39% of patients (≥70% stenosis in 7.3%) undergoing elective coronary angiography.[149] Furthermore, arterial stiffness may result in greater transmission of an elevated systemic blood pressure to glomerular capillaries and exacerbate glomerular hypertension. In one study, pulse wave velocity (PWV) and augmentation index (AI), markers of arterial stiffness, were identified as independent risk factors for progression to ESKD among subjects with CKD stage 4 or 5[150]; in another study, AI was an independent determinant of rate of creatinine clearance decline among subjects with CKD stage 3.[151] By contrast, neither PWV nor AI was a predictor of the rate of GFR decline in a cohort of subjects with CKD stages 2 to 4.[152] In two relatively small cohort studies of those with CKD, a diagnosis of CVD was associated with an increased risk of progression to ESKD[153,154] but, in the CRIC study, a history of any CVD at baseline was not associated with an increased risk of the primary endpoint of ESKD or 50% GFR reduction among 3939 participants. Conversely, in the same study, a history of heart failure was independently associated with a 29% higher risk of the primary outcome.[155] For further discussion of CVD in patients with CKD, see Chapter 54.

CONVENTIONAL BIOMARKERS

Biomarkers are parameters that are measured and evaluated to differentiate between normal and abnormal biological processes or predict adverse outcomes. For CKD, biomarkers may be considered a reflection of renal function impairment. Some of the standard biomarkers indicated in CKD progression include proteinuria and several routine serum biomarkers that are essential for evaluation of persons with reduced GFR.[99] The more novel biomarkers, which will be further discussed in Chapter 27, are serum and urinary substances that allow for more targeted risk prediction in CKD.[148]

PROTEINURIA

Proteinuria is an indicator of dysfunction in the glomerular filtration barrier and is therefore a marker of glomerulopathy and an index of disease severity. Proteinuria is often a result of glomerular injury and is associated with hyperfiltration, intraglomerular hypertension, glomerular hypertrophy, and glomerulosclerosis.[148] Experimental evidence has suggested that proteinuria may also contribute to progressive renal damage in CKD (see Chapter 30). A large body of evidence attests to a strong association between proteinuria and the risk of CKD progression, a relationship mediated through several mechanisms including tubular injury, inflammation, and fibrosis, as well as cardiovascular and all-cause mortality. Mass screening of a general population of 107,192 participants by dipstick urinalysis identified proteinuria as the most powerful predictor of ESKD risk over 10 years (OR, 14.9; 95% CI, 10.9 to 20.2).[20]

Similarly, among 12,866 middle-aged men enrolled in the Multiple Risk Factor Intervention Trial (MRFIT), proteinuria detected by dipstick test was associated with a significantly increased risk of developing ESKD over 25 years (HR for 1+ proteinuria, 3.1; 95% CI, 1.8 to 3.8; HR for ≥2+ proteinuria, 15.7; 95% CI, 10.3 to 23.9). Furthermore, detection of 2+ proteinuria or more increased the hazard ratio for ESKD associated with an eGFR < 60 mL/min/1.73 m^2 from 2.4 without proteinuria (95% CI, 1.5 to 3.8) to 41 with proteinuria (95% CI, 15.2 to 71.1).[156] Similar associations have been reported for measurements of urinary albumin in the general population. In the Nord-Trøndelag Health (HUNT 2) study, which included 65,589 adults, microalbuminuria and macroalbuminuria were independent predictors of ESKD after 10.3 years (HR, 13.0 and 47.2, respectively) and combining reduced eGFR with albuminuria substantially improved the prediction of ESKD.[157]

Among patients selected for having CKD from a wide variety of causes, baseline proteinuria has consistently predicted renal outcomes.[158–160] In three large prospective studies that included patients with nondiabetic CKD [MDRD study, Ramipril Efficacy In Nephropathy (REIN) study, and AASK], higher baseline proteinuria was strongly associated with a more rapid decline in GFR.[10,98,161,162] Similarly, among patients with diabetic nephropathy, the baseline urinary albumin-to-creatinine ratio (ACR) was a strong independent predictor of ESKD in the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) study and Irbesartan in Diabetic Nephropathy Trial (IDNT).[163,164] The findings of these individual studies have been confirmed by two large meta-analyses. In one analysis, which included nine general population cohorts (*N* = 845,125) and eight cohorts with increased risk of developing CKD (*N* = 173,892), urine ACRs of >30, 300, and 1000 mg/g were independently associated with progressive increases in the risk of ESKD, progressive CKD, and AKI, respectively (Fig. 20.4).[73] Among 21,688 patients known to have CKD from 13 studies, an eightfold higher urine ACR or protein-to-creatinine ratio (PCR) was associated with increased all-cause mortality (pooled HR, 1.40) and risk of ESKD (pooled HR, 3.04).[74] Further meta-analyses by the CKD Prognosis Consortium have shown that the magnitude of proteinuria remains a risk factor for ESKD independent of gender,[33] ethnicity,[44] age,[27] diabetes,[75] or hypertension.[76]

Recognition of the importance of proteinuria as a risk factor in CKD prompted the addition of an albuminuria category (A1 to A3) to the CKD classification proposed by KDIGO.[9] These important observations raise the question of how best to measure proteinuria. As discussed, all measurements of proteinuria have been reported to predict renal outcomes, including dipstick urinalysis, spot urine ACR, or PCR, ideally using the patient's first morning voids or, less frequently, using 24-hour urine collections. A secondary analysis of data from the RENAAL trial found that urine ACR measured on the first morning void was better than 24-hour urinary protein or albumin concentration as a predictor of time to doubling of the serum creatinine level or ESKD among patients with diabetes and CKD.[165] By contrast, retrospective analysis of data from 5586 patients with CKD reported similar HRs associated with urinary ACR and PCR for the outcomes of all-cause mortality, start of RRT, or doubling of serum creatinine.[166] Further analysis of these data identified a cohort of patients with a normal urine ACR, but elevated urine PCR in whom the risk of ESKD or death was intermediate between the groups with both urine ACR and PCR abnormal or normal.[167] Analysis of data from the CRIC study also reported that urine ACR and PCR had similar associations with complications of CKD.[168] Together, these data imply that any measure of proteinuria is better than no measurement. If the goal is to detect and monitor low levels of albuminuria (category A1 and A2), the urine ACR measured on the first morning void is best. For patients with CKD, urine ACR or PCR may be used, and there is some evidence that there may be added information gained by requesting both.[169]

SERUM ALBUMIN

Serum albumin levels are widely regarded as a marker of nutritional status but may also be reduced due to proteinuria or inflammation. Several studies have identified lower serum albumin levels as a risk factor for CKD progression. In the MDRD study, higher baseline serum albumin was associated with slower subsequent rate of GFR decline but, in a multivariable analysis, this was displaced by a similar correlation with baseline serum transferrin levels, another marker of protein nutrition.[10] Three studies have found associations between serum albumin and renal outcomes in patients with type 2 diabetes and CKD. Among 182 patients with a mean serum creatinine of 1.5 mg/dL at baseline, hypoalbuminemia was an independent risk factor for ESKD.[170] In a long-term follow-up of 343 patients, lower baseline serum albumin was an independent predictor of CKD progression and,[171] in the RENAAL study, lower serum albumin was an independent predictor of ESKD.[163] Similar observations have been reported in other forms of CKD. In a large cohort of patients with IgA nephropathy (N = 2269), lower serum total protein (composed largely of albumin) was an independent risk factor for ESKD.[172] In these studies, the predictive value of serum albumin was independent of and additional to that of proteinuria, indicating that it was not merely acting as a marker of albuminuria.

ANEMIA

Chronic anemia due to inherited hemoglobinopathy is associated with increased renal plasma flow, glomerular hyperfiltration, and subsequent development of proteinuria, hypertension, and ESKD.[173,174] Anemia is a common complication of CKD from any cause, and several studies have shown that it is also an independent predictor of CKD progression. In the RENAAL study, baseline hemoglobin was a significant independent predictor of ESKD among diabetic patients – each 1-g/dL decrease in hemoglobin was associated with an 11% increase in the risk of ESKD.[175] Baseline hemoglobin was also one of four variables included in the renal risk score developed from the RENAAL data.[163] Similarly, a higher hemoglobin level was independently associated with lower risk of progression to ESKD (halving of GFR or need for dialysis) or death among 131 patients with all forms of CKD (HR, 0.778; 95% CI, 0.639 to 0.948 for each 1-g/dL increase).[160] Furthermore, time-averaged hemoglobin of <12 g/dL was associated with a significantly increased risk of ESKD among 853 male veterans with CKD stages 3 to 5 (HR, 0.74; 95% CI, 0.65 to 0.84 for each 1-g/dL increase in hemoglobin).[176]

Consistent with the hypothesis that anemia contributes directly to CKD progression, two small randomized studies have reported a renoprotective benefit associated with erythropoietin therapy. Among patients with serum creatinine of 2 to 4 mg/dL and hematocrit < 30%, erythropoietin treatment was associated with significantly improved renal survival.[177] In nondiabetic patients with serum creatinine of 2 to 6 mg/dL, early treatment (started when hemoglobin < 11.6 g/dL) with erythropoietin alpha was associated with a 60% reduction in the risk of doubling the serum creatinine level, ESKD, or death versus delayed treatment (started when hemoglobin < 9.0 g/dL).[178] By contrast, two other studies that had left ventricular mass as their primary endpoint[179,180] and the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT)[181] found no effect of a high versus low hemoglobin target on the rate of decline in the GFR. Several studies have, however, reported adverse outcomes associated with normalization of hemoglobin in patients with CKD. In the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) study, randomization to a higher hemoglobin target (13 to 15 mg/dL) was associated with a shorter time to initiation of dialysis than a lower target (10.5 to 11.5 mg/dL).[182] In TREAT, randomization to a higher hemoglobin target was associated with an increased risk of stroke[181] and, in the Correction of Hemoglobin and Outcomes in Renal Insufficiency (CHOIR) study, a higher hemoglobin target was associated with an increased incidence of the combined endpoint of all-cause mortality, myocardial infarction, or hospitalization for congestive cardiac failure.[183]

DYSLIPIDEMIA

Lipid abnormalities are common in patients with CKD, and several studies have identified dyslipidemia as a susceptibility and progression factor for CKD. In population-based studies, several lipid profile abnormalities have been associated with an increased risk of developing CKD, including an elevated low-density lipoprotein (LDL)–to–high-density lipoprotein (HDL) cholesterol ratio,[184] higher triglyceride and lower HDL cholesterol levels,[185] lower HDL cholesterol levels[21] and elevated total cholesterol, low HDL cholesterol, and elevated total cholesterol-to-HDL cholesterol ratio.[186] In the MDRD study, lower HDL cholesterol levels independently predicted a more rapid decline in GFR[10]; in a smaller study of patients with CKD, total cholesterol, LDL cholesterol, and apolipoprotein B levels were all associated with a more rapid decline in the GFR.[187] Among 223 patients with IgA nephropathy, hypertriglyceridemia was independently predictive of CKD progression.[188] Hypercholesterolemia was reported to predict loss of renal function in patients with type 1 or 2 diabetes[189,190] and, among nondiabetic patients, CKD advanced more rapidly in those with hypercholesterolemia and hypertriglyceridemia.[191]

Randomized controlled trials of lipid lowering have produced mixed results with respect to renal outcomes. Subgroup analysis of a prospective randomized trial of pravastatin treatment in patients with previous myocardial infarction found that pravastatin slowed the rate of decline in patients with an eGFR < 40 mL/min/1.73 m^2, an effect that was also more pronounced in those with proteinuria.[192] Similarly, in the Heart Protection Study, patients with previous CVD or diabetes randomized to simvastatin treatment had a smaller increase in serum creatinine than those who received placebo.[193] In a placebo-controlled, open-label study, atorvastatin treatment in patients with CKD, proteinuria, and hypercholesterolemia was associated with preservation of creatinine clearance, whereas it declined in those receiving placebo.[194] By contrast, lipid lowering with fibrates was not associated with renoprotection in two studies,[184,195] although one study did show a reduced incidence of microalbuminuria in patients with type 2 diabetes receiving fenofibrate.[196]

Analysis of data from a relatively small subgroup of studies with renal endpoints recorded in a meta-analysis found that statin therapy was associated with a reduction in proteinuria but with no improvement in creatinine clearance in participants with CKD.[197] Furthermore, analysis of data from 3939 participants in the CRIC study found no association between total or LDL cholesterol and the risk of ESKD or 50% reduction in eGFR. Indeed, among participants with proteinuria of <0.2 g/day, higher LDL and total cholesterol were associated with a lower risk of reaching this endpoint.[198] The Study of Heart and Renal Protection (SHARP) is the largest randomized controlled trial to investigate the cardiovascular and renoprotective effects of lipid lowering in CKD. Patients with CKD or undergoing dialysis were randomized to treatment with simvastatin and ezetimibe or placebo. In 6245 participants with CKD not requiring dialysis, treatment resulted in an average reduction in LDL cholesterol of 0.96 mmol/L but was not associated with a reduction in the primary outcome of ESKD or secondary outcome of ESKD or creatinine doubling.[199] Together, evidence that dyslipidemia is a risk factor for CKD progression remains mixed, with the most recent studies indicating no association. Mechanisms whereby dyslipidemia may contribute to CKD progression are discussed in Chapter 51.

SERUM URIC ACID

Hyperuricemia is a common consequence of chronic renal failure and may also contribute to CKD progression. Most but not all population-based studies have identified hyperuricemia as an independent risk factor for the development of incident CKD. Similarly, most cohort studies that included people with CKD have identified a higher serum uric acid level as a risk factor for CKD progression. Possible mechanisms whereby hyperuricemia may contribute to CKD progression are exacerbation of glomerular hypertension,[200,201] endothelial dysfunction,[202,203] and proinflammatory effects.[204] By contrast, it is possible that an elevated uric acid concentration is acting as a marker of reduced kidney function or oxidative stress – uric acid is produced by xanthine oxidase, which also generates reactive oxygen species.

To date, only small studies investigating the effect of uric acid–lowering therapy on CKD progression have been published. A meta-analysis of eight trials found no difference in the eGFR among participants treated with allopurinol versus those with no treatment or placebo in five trials, whereas three trials that reported only serum creatinine reported benefit in favor of allopurinol. In five trials that measured proteinuria, no benefit was observed.[205] Together, published evidence suggests that an elevated serum uric acid level may act as a susceptibility and progression risk factor in CKD, but large randomized trials are still required to determine whether treatment of hyperuricemia is beneficial for slowing CKD progression.

METABOLIC ACIDOSIS

Metabolic acidosis commonly occurs in CKD patients and is the result of a decrease in functional nephron mass. As a result, there is a subsequent elevation of cortical ammonia levels, renin–angiotensin–aldosterone system activation, complement cascade activation, endothelial/aldosterone activation, inflammation, all of which lead to impaired renal acid clearance.[206] At least five studies have investigated metabolic acidosis as a risk factor in human CKD. In all except the MDRD study,[207] lower serum bicarbonate levels, even within the normal range, were independently associated with an increased risk of CKD progression.[208–211] Two small randomized trials have reported slowing of CKD progression with bicarbonate supplementation,[212,213] and another trial found that correction of acidosis with oral sodium bicarbonate or a diet rich in fruits and vegetables was associated with a lower rate of GFR decline.[214] Several randomized controlled trials have also indicated that correction of acidosis with bicarbonate supplementation has minimal sodium loading and impact on hypertension.[212,214] Bicarbonate supplementation is already recommended for patients with levels <22 mEq/L, but several studies are underway to investigate further whether it is beneficial in the setting of less severe acidosis.[215]

NOVEL BIOMARKERS

PLASMA ASYMMETRIC DIMETHYLARGININE

Asymmetric dimethylarginine (ADMA) is formed by the breakdown of arginine-methylated proteins and acts as an endogenous inhibitor of nitric oxide synthase to reduce nitric oxide production. The increased ADMA levels observed with a reduced GFR have been proposed as one mechanism for the endothelial dysfunction associated with CKD. Elevated ADMA levels are associated with CVD and cardiovascular mortality in patients with CKD.[216] In animal models, administration of ADMA was associated with the development of hypertension, increased deposition of collagen I and III and fibronectin in glomeruli and blood vessels, and rarefaction of peritubular capillaries.[217] Conversely, the overexpression of dimethylarginine dimethylaminohydrolase, the enzyme responsible for degradation of ADMA, was associated with reduced ADMA levels and amelioration of renal injury in rats after 5/6 nephrectomy, implying that ADMA may also promote CKD progression.[218]

Among 131 patients with CKD, a higher plasma ADMA level was an independent risk factor for ESKD or death (HR, 1.20; 95% CI, 1.07 to 1.35 for each 0.1-μmol/L increase).[160] In 227 relatively young patients with mild-to-moderate nondiabetic CKD, higher ADMA levels predicted progression to the combined endpoint of creatinine doubling or ESKD (HR, 1.47; 95% CI, 1.12 to 1.93 for each 0.1-μmol/L increase).[219] Finally, retrospective analysis of data from 109 patients with IgA nephropathy showed associations between ADMA levels and glomerular and tubulointerstitial injury. Furthermore, the plasma ADMA level was an independent determinant of annual GFR reduction rate.[220]

SERUM PHOSPHATE AND FGF23

When rats were fed a high-phosphate diet after uninephrectomy, renal calcium and phosphate deposition, as well as tubulointerstitial injury, were observed within 5 weeks.[221] Furthermore, in animals and humans with CKD, dietary phosphate restriction or treatment with oral phosphate binders was associated with reductions in proteinuria and glomerulosclerosis and attenuation of CKD progression.[222–225] Together, these data suggest that phosphate loading and/or hyperphosphatemia exacerbate renal injury in CKD. Three cohort studies of patients with CKD have identified higher serum phosphate levels as an independent risk factor for progression.[208,226,227] By contrast, the largest study to date, which included 10,672 participants with CKD, found no independent association between higher serum phosphate and risk of progression.[228] It should be noted, however, that the number of ESKD events was low, and the study therefore had limited power to detect a moderate association between serum phosphate levels and CKD progression.[229] In addition, higher levels of the phosphatonin, fibroblast growth factor 23 (FGF23), which is a hormone involved in calcium and phosphate regulation through decreasing serum phosphate and 1,25-dihydroxyvitamin D_3, have been identified as an independent predictor of CKD progression.[230,231] Along with adverse renal effects, higher levels of FGF23 are associated with CVEs and related surrogate outcomes in patients with CKD.[232–234] In the CRIC study, higher levels of FGF23 were determined to be an independent risk factor for ESKD, but only in patients with a higher baseline eGFR (HR, 1.3; 95% CI, 1.04–1.6 if the eGFR was 30–44 mL/min/1.73 m^2; and HR, 1.7; 95% CI, 1.1–2.4 if the eGFR was >45 mL/min/1.73 m^2).[232] A recent meta-analysis confirmed the association between elevated FGF23 and CVE as well as all-cause mortality but suggested that the relationship was not causal due to lack of an exposure–response relationship.[234] Currently, the exact clinical significance of FGF23 is uncertain but there appears to be potential for its use as an important biomarker, specifically in CKD and ESRD patients.[148]

NEUTROPHIL GELATINASE-ASSOCIATED LIPOCALIN

Neutrophil gelatinase-associated lipocalin (NGAL) is an iron-carrying protein expressed throughout the distal tubule epithelium and is overexpressed in response to AKI.[235] While it is known mostly for its role in AKI, the role of urinary NGAL in CKD progression is currently a point of interest in nephrology.[236] The Atherosclerosis Risk in Communities (ARIC) study found that baseline urinary NGAL was highest in those patients with incident CKD (OR, 2.1; 95% CI, 0.96–4.6).[237] In the CRIC study, urinary NGAL was determined to be an independent risk factor for ESRD (HR, 1.7; 95% CI, 1.2–2.5), but added no significant improvement in prediction of CKD progression beyond eGFR decline and proteinuria.[238]

KIDNEY INJURY MOLECULE-1

Kidney injury molecule-1 (KIM-1) is a transmembrane glycoprotein typically undetected in a normal kidney, but is detected in patients with AKI and CKD.[239] One cohort study found that serum KIM-1 levels increased with the stage of CKD, and were predictive of eGFR decrease and ESRD.[240] The Multi-Ethnic Study of Atherosclerosis determined that doubling of urinary KIM-1 levels was associated with CKD and rapid eGFR decrease >3 mL/min/1.73 m^2 per year (OR, 1.2; 95% CI, 1.02–1.3).[241]

SOLUBLE UROKINASE-TYPE PLASMINOGEN ACTIVATOR RECEPTOR

Soluble urokinase-type plasminogen activator receptor (suPAR) is a protein involved in cell adhesion and migration of endothelial and immune cells. Increased suPAR levels have been detected in patients with glomerular disease, specifically glomerulosclerosis, as well as adverse CVEs.[148] One study done by Hayek et al. using 2292 patients from the Emory Cardiovascular Bank found that higher baseline suPAR levels were associated with CKD (eGFR < 60 mL/min/1.73 m^2). This relationship was further validated using a sample of 347 patients in the Women's Interagency Human Immunodeficiency Virus Study, where again a higher baseline suPAR level was associated with greater incidence of CKD. Despite this, caution must be exercised because results may be misleading. For example, 30% of patients with an eGFR > 90 mL/min/1.73 m^2 also had an increase in baseline suPAR levels, indicating that levels may be linked to inflammation in addition to kidney function.[242] These results emphasize the need for further validation in other studies.

UROMODULIN

Uromodulin (UMOD) is a kidney-specific glycoprotein synthesized by epithelial cells found in the ascending loop of Henle. While its exact function is currently unknown, one proposed role has been in protection against urinary tract infections, innate immunity activation, and kidney stone prevention. GWASs indicated that single-nucleotide polymorphism (SNP) variants of the *UMOD* gene were associated with a decrease in eGFR and development of CKD in European cohorts.[243,244] One GWAS using 3203 Icelandic patients with CKD found an association between CKD and creatinine with a variant adjacent to the *UMOD* gene on chromosome 16p12, which was strengthened with age. For clinical purposes, however, there have been conflicting results with regard to the use of UMOD as a biomarker in CKD progression. A meta-analysis of GWAS for UMOD including six studies and 10,884 patients found that two loci located around the *UMOD* gene were associated with increased UMOD levels.[245] However, Shlipak et al. found that among 879 individuals of the Heart and Soul Study, UMOD SNP variants influenced UMOD levels, but did not affect incident CKD risk.[246] These conflicting results indicate that further investigation is required before clinical implementation of UMOD as a CKD biomarker.[148]

PROTEOMIC APPROACHES

Proteomics assess low-molecular-weight proteins using electrophoresis and mass spectrophotometry in both targeted (known pathophysiology) and nontargeted (unknown pathophysiology) approaches.[247] However, this approach poses several challenges given biosampling variability due to age, sex, diet, exercise, and other limitations. Using proteomics in a Scottish cohort with type II DM, investigators for the Innovative Diabetes Tools study were able to identify at least 62 of 207 serum protein biomarkers that were associated with rapid progression of CKD (>40% baseline eGFR loss in 3.5 years). The use of a 14-biomarker panel (including FGF23 and KIM-1) in risk prediction models allowed for an increase in the area under the receiver operating characteristic curve (AUROC) from 0.706 to 0.868.[248] Another study, which used capillary electrophoresis with mass spectrophotometry, looked at the utility of urinary proteomics in validating previously established biomarkers of CKD. The study used a general population cohort of 223 healthy individuals and 1767 patients with CKD (defined as eGFR < 90 mL/min/1.73 m^2 or urine albumin excretion > 30 mg/L), finding that the proteome performed better in detection and prediction of CKD progression, improving the AUROC based on baseline eGFR and albuminuria from 0.758 to 0.83.[249] The use of novel serum or urinary biomarkers will likely be of value in the future risk prediction models through enhancing discrimination beyond traditional risk predictors including GFR and proteinuria. Given the interplaying mechanisms that may result in any given patients with CKD, it is unlikely that any one single novel biomarker will have a large effect on risk prediction models. However, the development of biomarker panels through the use of proteomics shows potential in discriminating between disease and nondisease states as well as risk prediction, and its future role in the clinical setting is promising.[148]

OTHER BIOMARKERS

A number of other biomarkers are currently being investigated as risk factors in CKD. Although many have been reported to be associated with adverse outcomes, the challenge is to identify biomarkers that add to the predictive power of established risk factors. For a detailed review of biomarkers in kidney disease, see Chapter 27.

ENVIRONMENTAL RISK FACTORS

SMOKING

Population-based studies have identified cigarette smoking as an independent risk factor for various manifestations of CKD, including proteinuria,[250] elevated serum creatinine levels,[251] decreased eGFR,[21,252] and development of ESKD or death associated with CKD (HR, 2.6; 95% CI, 1.8 to 3.7).[28] In the latter study, 31% of the attributable risk of CKD was associated with smoking. In a longitudinal study of 10,118 middle-aged Japanese workers, smoking was associated with an increased risk of developing glomerular hyperfiltration (eGFR ≥ 117 mL/min/1.73 m^2; OR, 1.32 vs. nonsmokers) as well as proteinuria (OR, 1.51 vs. nonsmokers).[253] Two other similar longitudinal studies from Japan have confirmed that smoking is associated with an increased risk of developing proteinuria but with a higher mean eGFR than in nonsmokers.[254,255] In one study, smoking was associated with a reduced risk of developing CKD stage 3.[255] Smoking has been shown to increase the risk of progression of CKD due to diabetes,[256,257] hypertensive nephropathy,[258] glomerulonephritis,[259] lupus nephritis,[260] IgA nephropathy,[261] and adult PKD.[261] Randomized trials of the effect of smoking cessation on CKD progression are lacking but, in one observational study, smoking cessation was associated with less progression to macroalbuminuria and a slower rate of GFR decline than continued smoking in patients with diabetes.[262] Similarly, in the CRIC study, nonsmoking was associated with a reduced risk of CKD progression (HR, 0.68; 95% CI, 0.55 to 0.84), atherosclerotic CVEs (HR, 0.55; 95% CI, 0.40 to 0.75), and mortality (HR, 0.45; 95% CI, 0.34 to 0.60).[263] Among 9270 participants with CKD in the SHARP, current smoking was associated with an increased risk of all-cause mortality, CVE, and cancer but was not associated with risk of ESRD or rate of GFR decline in 6245 participants not receiving dialysis at baseline.[264] Possible mechanisms whereby cigarette smoking may contribute to renal damage include sympathetic nervous system activation, glomerular capillary hypertension, endothelial cell injury, and direct tubulotoxocity.[265]

ALCOHOL

The role of alcohol consumption as a potential risk factor for CKD remains unclear. One case–control study found a significant association between ESKD and consumption of more than two alcoholic drinks daily,[266] whereas another similar study found no association (with the exception of moonshine).[267] Some population-based studies have found that alcohol consumption is not related to CKD risk,[268–270] but one study found a significant association of heavy alcohol intake (more than four drinks daily) and prevalent CKD, as well as the risk of developing CKD in participants with a normal GFR.[252] Furthermore, heavy alcohol intake substantially increased the risk of CKD progression associated with smoking, such that participants who smoked and drank heavily had an almost fivefold increased risk of developing CKD.[252]

Conversely, several large cohort studies have reported an inverse relationship between alcohol consumption and the risk of developing CKD[271,272] or ESKD.[273] Another study found that moderate-to-heavy alcohol consumption was associated with an increased risk of developing albuminuria but decreased risk of eGFR < 60 mL/min/1.73 m^2.[274] The most rigorous study published to date investigated the incidence of CKD defined by an eGFR determined using the combined cystatin C and creatinine equation or albuminuria > 30 mg/day based on two consecutive 24-hour urine collections. The risk of developing CKD over a mean of 10.2 years decreased progressively with increasing alcohol consumption: HR of 0.85 (95% CI, 0.69 to 1.04) for occasional alcohol consumption (<10 g/week), HR of 0.82 (95% CI, 0.69 to 0.98) for light alcohol consumption (10 to 69.9 g/week), HR of 0.71 (95% CI, 0.58 to 0.88) for moderate alcohol consumption (70 to 210 g/week), and HR of 0.60 (95% CI, 0.42 to 0.86) for heavier alcohol consumption (>210 g/week).[275]

RECREATIONAL DRUGS

The role of recreational drugs as a risk factor for CKD has not been widely studied, but one case–control study reported a positive association between heroin, cocaine, or psychedelic drug use and ESKD.[276] Following reports of a specific renal lesion characterized by proteinuria and FSGS, termed "heroin nephropathy," other investigators reported a wide range of renal lesions in patients with a history of heroin abuse. It is unclear whether the observed renal lesions resulted from direct effects of heroin or were attributable to impurities in the drug or associated blood-borne virus infections and endocarditis. An association with renal amyloidosis, possibly due to chronic skin infections, has also been reported.[277] Interestingly, heroin abuse was not associated with an increased risk of mild CKD in 647 hypertensive patients who showed an association between illicit drug abuse and CKD.[278] Cocaine exerts several adverse effects that may induce renal injury, including rhabdomyolysis, vasoconstriction, activation of the renin–angiotensin–aldosterone system, oxidative stress, and increased collagen synthesis.[277] Furthermore, chronic administration of cocaine to rats resulted in multiple renal lesions, including glomerular atrophy and sclerosis, tubule cell necrosis, and areas of interstitial necrosis.[279] Among 647 patients attending a hypertension clinic, a history of any illicit drug use was independently associated with a relative risk of 2.3 (95% CI, 1.0 to 5.1) for mild CKD, whereas cocaine and psychedelic drug use were associated with relative risks of 3.0 (95% CI, 1.1 to 8.0) and 3.9 (95% CI, 1.1 to 14.4), respectively.[278] In one prospective cohort study using 2286 participants of the Life Span study, use of opiates and cocaine were both found to have an association with greater OR of reduced eGFR (<60 mL/min/1.73 m^2; OR, 2.71 and 95% CI, 1.50–4.89 for opiates; OR, 1.40 and 95% CI, 0.87–2.24 for cocaine) and were associated with greater OR of albuminuria (ACR > 30 mg/g; OR, 1.20 and 95% CI, 0.83–1.73 for opiates; OR, 1.80 and 95% CI, 1.29–2.51 for cocaine).[280]

ANALGESICS

Analgesic nephropathy has been well described as a cause of CKD and ESKD resulting from abuse of combination analgesics containing aspirin and phenacetin; this was prevalent in Australia and Switzerland until the sale of these products was restricted[281] (see Chapter 81). Cohort studies of participants without CKD at baseline have, however, not reported strong associations between analgesic use and the development of CKD. Among 1697 women in the Nurses Health Study, consumption of >3000 g of acetaminophen was associated with an increased risk of a GFR decline of >30 mL/min/1.73 m^2 over 11 years (HR, 2.04; 95% CI, 1.28 to 3.24), but greater use of aspirin or nonsteroidal antiinflammatory drugs (NSAIDs) was not associated with increased risk.[282] Among 4494 male physicians, there was no association between occasional-to-moderate use of aspirin, acetaminophen, and NSAIDs and GFR decline over 14 years.[283] By contrast, analgesic use may exacerbate the progression of established CKD. In one large study of 19,163 patients with newly diagnosed CKD, use of aspirin, acetaminophen, or NSAIDs was associated with an increased risk of progression to ESKD in a dose-dependent manner. Among cyclooxygenase 2 inhibitors, use of rofecoxib but not celecoxib was associated with increased risk of ESKD.[284] In a cohort study of 4101 people with rheumatoid arthritis, chronic use of NSAIDs was not associated with a more rapid GFR decline in the entire study population, but NSAID use was independently associated with more rapid GFR decline in a small minority of patients with CKD stage 4 or 5 ($N = 17$).[285] A meta-analysis of three studies that included data from 54,663 participants with CKD stages 3 to 5 found no association between regular NSAID use and accelerated GFR decline (defined as ≥15 mL/min over 2 years) but did report an association with high-dose NSAID use (defined as 90th percentile or above in one study and not defined in the other) in two studies.[286] The use of single-compound acetaminophen or aspirin was reported not to accelerate progression among patients with CKD stage 4 or 5,[287] but a systematic review of the safety of paracetamol treatment reported an increased risk of renal adverse events in three of four observational studies – in addition to an increased risk of all-cause mortality and cardiovascular and gastrointestinal adverse events in other studies.[288]

As the authors noted, these results must be interpreted with some caution due to the possibility of "confounding by indication" resulting from associations between the indication for prescribing analgesia and adverse outcomes.

HEAT STRESS

An epidemic of CKD has been reported in Central American farmworkers, fittingly termed Mesoamerican nephropathy (MeN). The disease itself has been most prominent in El Salvadorian sugarcane cutters, but has been reported in areas ranging from southern Mexico to Costa Rica. MeN is usually asymptomatic and is not associated with proteinuria and the usual CKD-related comorbidities, such as diabetes or hypertension, but it does have a high risk of progressing to ESRD. While the exact pathology causing MeN has been debated, varying from side effects of pesticides to medication use, the currently accepted hypothesis is that it is linked to heat-induced dehydration.[289–291] A recent population-based study done by García-Trabanino et al. looked at the renal effects of dehydration after a single shift in the sugarcane fields of El Salvador.[291a] Using 168 male sugarcane cutters, aged 18–49 years, who worked a mean (SD) shift time of 4 (1.4–11) hours, they found significant changes between preshift and postshift assessments of renal biomarkers. Specifically, there was a consistent increase in mean urine specific gravity, urine osmolality, and creatinine, and a decrease in urinary pH. The study also looked at the prevalence of reduced eGFR (<60 mL/min/1.73 m^2) in the sample, noting that 14% of these workers had signs of potential CKD. Recent findings have also indicated that the Mesoamerican demographic in which MeN is occurring also has a higher exposure to nephrotoxins, high fructose intake, and higher use of NSAIDs than other populations, a combination of which may further exacerbate the risk of CKD.[292] CKD with similar characteristics has been identified in other tropical areas of the world and an alternative term, CKD of unknown etiology (CKDu), has been proposed.[293]

HEAVY METALS

Chronic exposure and subsequent toxicity of heavy metals, specifically lead and cadmium, has been known to result in well-recognized instances of nephropathy. Overt lead toxicity results in the well-recognized entity of lead nephropathy, characterized by chronic interstitial nephritis and an association with gout. In addition, epidemiologic studies have reported that mild elevations in blood lead levels are associated with moderate reductions in GFR and/or hypertension in the general population.[294,295] Furthermore, a prospective study has identified elevations in blood lead levels and body lead burden (BLB) within the normal range as important risk factors for progression in patients with CKD.[296] Similarly, BLB was a risk factor for progression among 108 patients with low-normal BLB values and no history of lead exposure.[297] Furthermore, randomization to chelation therapy was associated with a modest improvement in GFR over 24 months versus a small decline in those randomized to control (+6.6 ± 10.7 versus −4.6 ± 4.3 mL/min/1.73 m^2; $P < .001$).[297]

Like lead, chronic exposure to cadmium is also associated with a distinctive nephropathy, which is characterized by proximal tubule damage and low-molecular-weight proteinuria.[298] Furthermore, low-level cadmium exposure resulting from environmental contamination was associated with tubular proteinuria[299]; and analysis of data from 14,778 participants in NHANES showed an independent increased risk of albuminuria, reduced GFR, or both between the highest and lowest quartiles of blood cadmium levels.[300] Comparison of the lowest and highest quartiles for blood cadmium and lead levels showed an even greater increased risk of albuminuria, reduced GFR, or both.[300] In another NHANES study, blood and urine cadmium levels were positively correlated with the urine ACR and negatively associated with GFR. Higher blood and urine cadmium levels were independently associated with albuminuria, and higher blood cadmium levels were associated with albuminuria and a reduced GFR.[298] Occupational or low-level environmental exposure to cadmium was associated with an increased risk of ESKD in a population-based study from Sweden.[301]

RENAL RISK SCORES

The focus on investigating risk factors that predict the development and/or progression of CKD in diverse populations has led to the observation that a relatively small group of risk factors appears to be common to different forms of CKD. This supports the notion of a common pathway of mechanisms that underlie the progression of CKD. It has also led to the proposal that these common risk factors could be combined to develop a renal risk score to predict the development and future risk of progression of CKD in a manner analogous to the Framingham risk score for predicting cardiovascular risk in the general population.[302] The revised classification system for CKD proposed by KDIGO has in part addressed this need by incorporating the evidence that a reduced GFR and albuminuria are powerful risk factors to incorporate into a system in which CKD categories correspond to risk categories.[9]

In addition, considerable progress has been made in developing renal risk scores to facilitate more accurate risk prediction. These may conveniently be divided into two groups – those that apply to the general population (i.e., without CKD as baseline) and those that predict the risk of progression in patients already diagnosed with CKD. In addition, one study has developed a risk score to predict the development of CKD after an episode of AKI.

METHODS OF RENAL RISK SCORE DEVELOPMENT

Prior to clinical implementation, renal risk scores must first hold both internal and external validity, improvement over the current risk classification system, and must be easily integrated into clinical setting. The presence of internal validity indicates that the prediction model has been developed from a sample that accurately reflects the relationships between the variables used in the prediction model and the outcome of interest. These predictor variables must have clear definitions and must be measured in each patient well before the final outcome is reached. The outcome itself, in order to avoid bias, must also be clearly defined and assessed equally in all patients, blind to the status of the predictor variable.[208] In order to develop such a risk model, appropriate statistical approaches must be taken. If censoring of the variable is negligible and the follow-up period is clearly defined, logistic regression is used. However, if there is significant censoring present, the Cox proportional hazards model is preferred.[303,304] In addition to the general validation issues discussed earlier, several metrics specific to performance of the prediction model are used in order to assess internal validity.

METRICS OF MODEL PERFORMANCE

DISCRIMINATION

Discrimination is used as a method in determining the model's ability to accurately assign higher probabilities to patients in whom the event of interest occurs, compared with those in whom it does not. The concordance or C-statistic, which is the most commonly used tool of discrimination, is defined as the proportion of times the prediction model correctly discriminates between a randomly selected pair of individuals (case and control), and can be considered identical to the "area under the receiver operator curve" (AUROC). Like the AUROC, a C-statistic of 0.50 indicates that the model performance is essentially no better than chance; a C-statistic of 0.70–0.80 indicates that the model has a good performance in discrimination; and a C-statistic of >0.80 indicates excellent discriminatory performance. The comparison of C-statistics is a frequently used method for comparing the discrimination of multiple prediction models in order to determine which one is superior in risk prediction).[305–308] However, as the value of the C-statistic approaches 1, it becomes difficult to appreciate a significant difference in the values of different models and a more sensitive method, such as the Integrated Discrimination Index (IDI), must be used. The IDI looks at the difference in the discrimination slopes of the two separate models and describes it on an absolute and relative scale, achieving an effective comparison of discrimination values when the C-statistics are no longer sufficient.[309,310]

CALIBRATION

Model calibration is another metric of model performance, referring to how well the predictions made from the model align with actual data. For logistic regression models, the Hosmer–Lemeshow chi square statistic is the most commonly used method for such assessments. This method ranks participants based on predicted probability into deciles and the mean probability of each decile is then compared with the actual frequency of outcomes among participants in each decile. A chi square is then used to assess for significant discrepancies between the predicted and actual probabilities. Chi square statistics that are said to be significant indicate that the model calibration is suboptimal.[311] The metric of calibration is important for clinical risk prediction models because a poorly calibrated model will result in either underprediction or overprediction of risk, which is problematic when clinical decision-making is dependent on such prediction models.[208]

RECLASSIFICATION

It is standard for clinical treatments and tests to be selected based on the predicted risk of having an event. In the development of a new prediction model, it is essential to consider whether or not it reclassifies patients into more appropriate risk categories than was the case in the old prediction model. If a patient has an event and the new model had assigned the said patient to an appropriately higher-risk category, the new model will be considered a success. Similarly, for patients who do not have an event, the new model is successful if it reclassifies the patient to lower-risk category than the previous model. However, if the new prediction model reassigns the patient to a risk category that is opposite of the actual outcome, the new model is considered unsuccessful and is not superior to the previous model. This success or failure can be quantified by the Net Reclassification Index, values for which range from –2.0 to +2.0, with positive values indicating successful reclassification and negative values indicating unsuccessful reclassification).[309,310]

CLINICAL UTILITY

Even the best risk prediction model is unusable in a clinical setting if it cannot be rapidly and efficiently implemented. As previously indicated, prediction models are derived through a series of complex logistic or proportional hazards models that cannot be easily applied to patients using simple calculations. This means that the model, once validated, needs to be translated into a simpler clinically useful bedside tool. However, in simplifying the scoring system, there is some inevitable loss of precision, discrimination, and calibration. Recently, the advent of rapid access to Web-based calculators or smartphone apps has allowed for complex prediction models to be applied in their original form via a simplified user interface, and without the need for complex calculations by the user.[312–315]

EXTERNAL VALIDITY

External validity, unlike internal validity, addresses whether or not the results of the study sample can be applied to the general population. The external validity of a prediction model can never be assumed, because it is likely that a model generated from one set of data (derivation cohort) will not perform exactly the same in other cohorts. This is likely due to underlying biological factors, including differences in disease or physiology of the cohorts and false associations between predictors and outcomes in the original derivation cohort. Attempts to minimize errors between cohorts can be achieved by careful selection of cohorts that are representative of the clinical condition, and by choosing predictor variables based on clinical relevance rather than statistical associations.[308]

MODELS PREDICTING INCIDENT CHRONIC KIDNEY DISEASE

Risk scores have been proposed to assess the risk of developing CKD in the general population and, in some cases, its subsequent progression. These are summarized in Table 20.3 and have been assessed in a systematic review.[316] In the first study, data from 8530 adults included in NHANES were used to identify risk factors for prevalent CKD (defined as eGFR < 60 mL/min/1.73 m^2). The authors proposed a risk score that included age, female gender, hypertension, anemia, diabetes, peripheral vascular disease, history of CVD, congestive heart failure, and proteinuria. The AUROC was high at 0.88, and a score of ≥4 resulted in a sensitivity of 92% and specificity of 68%. The positive predictive value was low at 18%, but the negative predictive value was 99%. External validation using data from the ARIC study gave an AUROC value of 0.71.[317] This was a cross-sectional study, and the risk score therefore did not predict the risk of future CKD. Rather, it identified individuals at increased risk of having current undiagnosed CKD. As such, it would be useful for guiding efforts to screen populations for CKD, but gives no information about the future risk of CKD progression. The applicability of the score to general populations is somewhat weakened by the inclusion of two variables that require prior laboratory testing: anemia and proteinuria. Furthermore, the presence of significant proteinuria is sufficient to diagnose CKD in the absence of any reduction in GFR.

Another risk score was developed to predict the risk of incident CKD using combined data from 14,155 participants in the ARIC study and Cardiovascular Health Study (CHS; ≥45 years) with baseline eGFR > 60 mL/min/1.73 m^2. After identifying 10 predictors of incident CKD (defined as eGFR < 60 mL/min/1.73 m^2 during follow-up for up to 9 years), they proposed a simplified model based on eight variables: age, anemia, female gender, hypertension, DM, peripheral vascular disease, and history of congestive heart failure or CVD. This gave an AUROC value of 0.69, and a score of ≥3 resulted in a sensitivity of 69% and specificity of 58% but the positive predictive value was low, only 17%.[318] A similar study used data from 2490 participants in the Framingham Heart Study to produce a risk score for incident CKD, defined as eGFR < 60 mL/min/1.73 m^2. The final model included age, diabetes, hypertension, baseline eGFR, and albuminuria and gave an AUC value of 0.813. External validation was performed with data from the ARIC study (AUROC = 0.79 and 0.75 in whites and blacks, respectively).[319] One further study developed a risk score for incident CKD (eGFR < 60 mL/min/1.73 m^2) in 5168 Chinese participants. Age, BMI, diastolic blood pressure, type 2 diabetes, previous stroke, serum uric acid, postprandial blood glucose, hemoglobin A_{1c} levels, and proteinuria > 100 mg/dL were included in two risk scores (one using clinical variables only and a second with all variables) that gave an AUROC value of 0.77. The study was limited by relatively short follow-up (median, 2.2 years) and a low AUROC value of 0.67 for external validation data.[320]

These scores are useful to identify individuals at higher risk of developing CKD for monitoring or intervention to reduce risk, but do not distinguish the minority who are at risk of progressing to ESKD from the majority who are at low risk. In an attempt to identify only high-risk individuals, another group used data from persons (775,091 women and 799,658 men aged 35 to 74 years, without a recorded diagnosis of CKD) in 368 primary care practices in the United Kingdom to develop a risk score. Two outcomes were studied over a period of up to 7 years – moderate-to-severe CKD (defined as kidney transplantation, dialysis, diagnosis of nephropathy, proteinuria, or eGFR < 45 mL/min/1.73 m^2) and ESKD (defined as kidney transplantation, dialysis, or eGFR < 15 mL/min/1.73 m^2); separate risk scores were developed for men and women. The final model for moderate-to-severe CKD included age, ethnicity, social deprivation, smoking, BMI, systolic blood pressure, diabetes, rheumatoid arthritis, CVD, treated hypertension, congestive cardiac failure, peripheral vascular disease, use of NSAIDs, and family history of kidney disease. In women, it also included systemic lupus erythematosus and history of kidney stones. The model for ESKD was similar but did not include NSAID use. Internal and external validation was performed, giving AUROC values of 0.818 to 0.878.[321]

One important limitation of this study is that it was observational and therefore likely to be subject to significant

Table 20.3 Renal Risk Scores for the General Population

	Study					
Parameter	**SCORED**[317]	**SCORED2**[318]	**Chinese**[320]	**Framingham**[319]	**QKIDNEY**[321]	**PREVEND**[322]
Population	NHANES	CHS + ARIC	General population	FHS	QResearch	eGFR > 45
Outcome	eGFR < 60 mL/min/1.73 m^2 (prevalent)	eGFR < 60 mL/min/1.73 m^2 (incident)	eGFR < 60 mL/min/1.73 m^2 (incident)	eGFR < 60 mL/min/1.73 m^2 (incident)	CKD, ESKD	Rapid ↓ GFR
Factor	Age	Age	Age	Age	Age	Age
	Female	Female				
					Ethnicity	
					Deprivation	
					Family history	
					Smoking	
			BMI			
	HT	HT		HT	HT	HT
	DM	DM	Type 2 DM	DM	DM	
	PVD	PVD			PVD	
	CVD	CVD	Stroke		CVD	
	CCF	CCF			CCF	
					RA	
	Anemia	Anemia				
			DBP		SBP	SBP
					BMI	
				eGFR		eGFR
	Proteinuria		Proteinuria	Albuminuria		Albuminuria
						CRP
			Uric acid			
			HbA_{1c}			
			Glucose			
					NSAIDs	
AUC	0.88	0.69	0.77	0.81	0.88	0.84
Validation	ARIC	CHS + ARIC	General population	ARIC	THIN	Internal

ARIC, Atherosclerosis Risk in Communities; *AUC,* area under the concentration–time curve; *BMI,* body mass index; *CCF,* congestive cardiac failure; *CHS,* Cardiovascular Health Study; *CKD,* chronic kidney disease; *CRP,* C-reactive protein; *CVD,* cardiovascular disease; *DBP,* diastolic blood pressure; *DM,* diabetes mellitus; *eGFR,* estimated glomerular filtration rate; *ESKD,* end-stage kidney disease; *FHS,* Framingham Heart Study; *HbA_{1c},* hemoglobin A_{1c}; *HT,* hypertension; *NHANES,* National Health and Nutrition Examination Survey; *NSAIDs,* nonsteroidal antiinflammatory drugs; *PREVEND,* Prevention of Renal and Vascular End-stage Disease; *PVD,* peripheral vascular disease; *RA,* rheumatoid arthritis; *SBP,* systolic blood pressure; *THIN,* the Health Improvement Network.

bias. Furthermore, only 56% of participants had a serum creatinine level recorded at inclusion, and it is therefore probable that several had undiagnosed CKD. The composite outcome of moderate-to-severe CKD was composed of several disparate variables and is therefore not clinically useful, but the ESKD outcome is relevant because it identified only the minority at increased risk of severe progression. This study also illustrates the utility of a risk score that could be programmed into primary care computer systems to alert family practitioners to patients who are at risk of progression to ESKD.

Another study used data from 6809 participants in the Prevention of Renal and Vascular End-stage Disease (PREVEND) study to develop a risk score with the primary outcome of progressive CKD over 6.4 years, defined as the 20% of participants with the most rapid decline in GFR and eGFR < 60 mL/min/1.73 m^2. The final risk score included baseline eGFR, age, albuminuria, systolic blood pressure, C-reactive protein, and known hypertension. The AUROC value was 0.84, and internal validation was performed using a bootstrapping procedure.[322] Despite this, the risk score has a relatively low sensitivity and positive predictive value. The proposed threshold score of ≥27 identified 2.1% of the population as high risk, but with a sensitivity of only 15.7% and a positive predictive value of 28.1%. The specificity and negative predictive value were high at 98.4% and 96.7%, respectively. Thus a low score is useful to identify low-risk individuals but a high score does not identify most high-risk individuals. Selecting a lower threshold would improve sensitivity with some reduction in specificity and could be used to identify a group at intermediate risk for closer monitoring. Limitations of this study are that it was performed in a white population and was not validated externally. External validation in other populations is therefore required before it can be considered for clinical use.

MODELS PREDICTING KIDNEY FAILURE

Several risk scores have been developed for patients with diagnosed CKD in a variety of study populations and are summarized in Table 20.4 and in systematic reviews.[316,323]

Table 20.4 Renal Risk Scores for Patients With Chronic Kidney Disease (CKD)

	Study					
Parameter	**RENAAL[163]**	**AIPRD[324]**	**IGAN[172]**	**KPC[325]**	**CRIB[226]**	**SHC[208]**
Disease studied	DN	CKD	IgAN	CKD, stage 3 or 4	CKD, stages 3 to 5	CKD, stages 3 to 5
Variables		Age	Age	Age		Age
			Male	Male	Female	Male
	Creatinine	Creatinine	1/creatinine	eGFR	Creatinine	eGFR
	UACR	UPE	Proteinuria	N/A	UACR	UACR
		SBP	SBP	HT		
				DM		
	Alb		TP			Alb
	Hb			Anemia		
						Calcium
					Phos	Phos
						Bicarb
			Histol			
			Hematuria			
Outcome	ESKD	ESKD or doubling of serum creatinine level	ESKD	RRT	ESKD	ESKD
AUC			0.939	0.89	0.873	0.917
Validation	No	No	No	No	Yes	Yes

AIPRD, ACE Inhibition in Progressive Renal Disease study; *Alb,* serum albumin; *AUC,* area under the concentration–time curve; *Bicarb,* serum bicarbonate; *CRIB,* Chronic Renal Impairment in Birmingham study; *DM,* diabetes mellitus; *DN,* diabetic nephropathy; *eGFR,* estimated glomerular filtration rate; *ESKD,* end-stage kidney disease; *Hb,* hemoglobin; *Histol,* histologic grade; *HT,* hypertension; *IgAN,* IgA nephropathy; *KPC,* Kaiser Permanente Cohort; *N/A,* not available; *Phos,* serum phosphate; *Proteinuria,* urine dipstick proteinuria; *RENAAL,* Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan study; *RRT,* renal replacement therapy; *SBP,* systolic blood pressure; *SHC,* Sunnybrook Hospital cohort; *TP,* serum total protein; *UACR,* urine albumin-to-creatinine ratio; *UPE,* 24-hour urinary protein excretion.

Analysis of data from 1513 patients with diabetic nephropathy included in the RENAAL study identified urine ACR, serum albumin, serum creatinine, and hemoglobin as independent risk factors for ESKD. A risk score was derived from the coefficients of these variables in the Cox proportional hazards model, which successfully separated the participants into quartiles of ESKD risk (Fig. 20.5), with a marked difference in risk between the first and last quartiles (6.7 vs. 257.2/1000 patient-years).[163]

Among 1860 patients with nondiabetic CKD from a combined database of 11 clinical trials, Cox proportional hazards analysis identified age, serum creatinine, proteinuria, and systolic blood pressure as independent risk factors for the combined endpoint of time to ESKD or creatinine doubling. Using similar methodology as the previous study, a risk model based on these variables was developed to stratify patients into quartiles of risk. The annual incidence of the combined endpoint was 0.4% versus 28.7% in the lowest versus highest quartile for patients in the control group and 0.2% versus 19.7% in those randomized to angiotensin-converting enzyme inhibitor treatment.[324] Analysis of data from 2269 patients with IgA nephropathy identified systolic blood pressure, proteinuria (assessed with urine dipstick test), serum total protein, 1/serum creatinine, and histologic grade at initial biopsy as predictors of time to ESKD. Age, gender, and severity of hematuria were added to these variables to develop a scoring system for estimating 4- and 7-year cumulative incidence of ESKD. There was close agreement between estimated and observed risks (AUROC value = 0.939).[172]

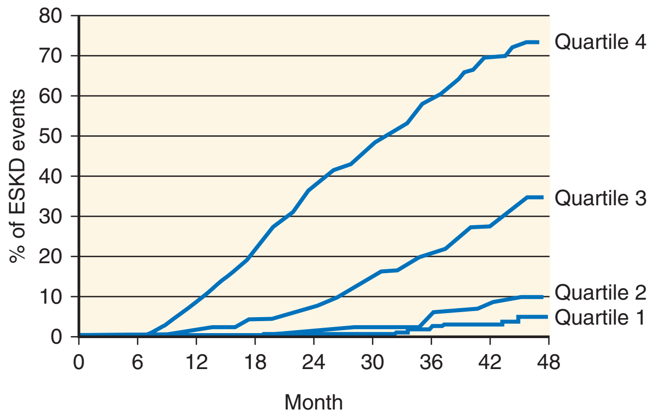

Fig. 20.5 Kaplan–Meier curve for the end-stage kidney disease (ESKD) endpoint stratified by quartile of risk score in 1513 patients with diabetic nephropathy from the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) study. *NIDDM,* Non–insulin-dependent diabetes mellitus. (From Keane WF, Zhang Z, Lyle PA, et al. Risk scores for predicting outcomes in patients with type 2 diabetes and nephropathy: the RENAAL study. *Clin J Am Soc Nephrol.* 2006;1:761–767.)

In a retrospective study, data from 9782 patients with CKD stages 3 to 4 were analyzed with regard to a primary outcome of onset of RRT (dialysis or transplantation). Six independent risk factors were identified—age, male gender, eGFR, hypertension, diabetes, and anemia—and were incorporated into a risk score that stratified participants into quintiles of risk.

The risk of progression to RRT was 19% in the highest risk quintile versus 0.2% in the lowest. The AUROC value was 0.89, and observed risk differed from predicted by <1%.[325] One limitation of this study, apart from its retrospective design, was the lack of data regarding proteinuria.

Among 382 patients with CKD stages 3 to 5 from the Chronic Renal Impairment in Birmingham (CRIB) study, independent risk factors for progression to ESKD during a mean of 4.1 years' follow-up were female gender, serum creatinine, serum phosphate, and urine ACR. A risk score was derived from these variables (AUROC value = 0.873) and externally validated in a similar cohort of patients with CKD stages 3 to 5 (East Kent cohort), giving an AUROC value of 0.91, even though the urine ACR was not available in the validation cohort.[226]

KIDNEY FAILURE RISK EQUATIONS

The kidney failure risk equations (KFREs) for predicting ESKD were developed by Tangri and colleagues using data from a cohort of 3449 Canadian patients with CKD stages 3 to 5 (Sunnybrook Hospital cohort).[208] It included age, male gender, eGFR, albuminuria, serum calcium, serum phosphate, serum bicarbonate, and serum albumin (AUROC value = 0.917). External validation was performed with data from a separate cohort of 4942 patients with CKD stages 3 to 5 (British Columbia CKD Registry), giving an AUROC value of 0.841. There was close agreement between predicted and observed risk in the validation cohort (Fig. 20.6). Simpler three-variable (age, gender, and eGFR) and four-variable models (age, gender, eGFR, and albuminuria) also performed well [C-statistic (equivalent to AUROC) of 0.89 and 0.91, respectively, in the development cohort; Table 20.5], but the eight-variable equation (C statistics 0.92) performed better in the validation cohort with respect to discrimination (C-statistic, 0.79 and 0.83, respectively, for the three- and four-variable equations), calibration, and reclassification. The authors facilitated clinical application of this risk score by producing an electronic risk calculator and a smartphone

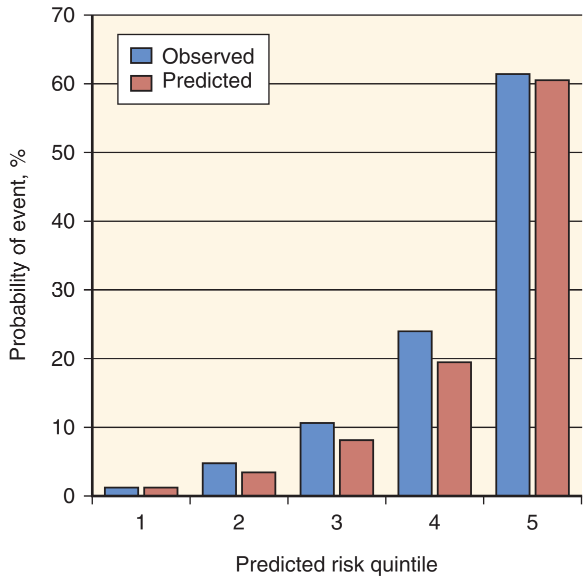

Fig. 20.6 Renal risk score predicted versus observed risk of developing end-stage kidney disease at 3 years in a validation cohort of patients with chronic kidney disease stages 3 to 5. (From Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA* 2011;305:1553–1559.)

Table 20.5 Hazard Ratios and Goodness of Fit for Sequential Models in the Development Data Set for Three-, Four-, and Eight-Variable Kidney Failure Risk Equations

Variable	Three-Variable Model	Four-Variable Model	Eight-Variable Model
Hazard Ratio for Kidney Failure (Censored for Mortality)			
Baseline GFR, per 5 mL/min/1.73 m^2	0.54	0.57	0.61
Age, per 10 year	0.75	0.8	0.82
Male sex	1.46	1.26	1.16
Log spot urine ACR[a]		1.60	1.42
Serum albumin, per 0.5 g/dL			0.84
Serum phosphate, per 1.0 mg/dL			1.27
Serum bicarbonate, per 1.0 mEq/L			0.92
Serum calcium, per mg/dL			0.81
Goodness of Fit Metrics			
C-statistics	0.89	0.91	0.92
Akaike information criterion[b]	4834	4520	4432
P value[c]	<.001	<.001	<.001

[a]Hazard ratio for ACR represents a 1.0 point higher ACR on the natural log scale. For the average patient with 20 mg/g of albuminuria, this represents an increase to 55 mg/g.
[b]Null values for C-statistic and Akaike information criterion are 0.50 and 5569, respectively. Higher values for C-statistic and lower values for Akaike information criterion indicate better models.
[c]*P* values are for comparison of C-statistics between sequential models.
ACR, Albumin-to-creatinine ratio; *GFR,* glomerular filtration rate.
From Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA* 2011;305:1553–1559.

app (available for no charge at www.qxmd.com/calculate-online/nephrology/kidney-failure-risk-equation) that report an estimated risk of ESKD at 2 and 5 years. Further external validation of this risk equation has been reported by independent investigators from the MASTERPLAN (Multifactorial Approach and Superior Treatment Efficacy in Renal Patients with the Aid of Nurse practitioners) study, a cohort of 595 people from the Netherlands with CKD stages 3 to 5. The model again performed well with the eight-variable equation, giving an AUROC value of 0.89.[326] A further advantage of the Tangri risk score is that it includes variables that are all available to a biochemistry laboratory, making it possible for laboratories to automate reporting of a risk score with each eGFR.

In order to determine if KFREs could be implemented clinically on a global scale, a multinational validation study of the equations was performed in 35 cohorts consisting of 720,000 patients with CKD (stages 3–5) from 23 countries across four continents.[327] After a 4-year follow-up, 23,829 cases of kidney failure were observed in the cohorts. The KFRE was able to achieve excellent discrimination through all cohorts (C-statistic, 0.90; 95% CI, 0.89–0.92 at 2 years; C-statistic, 0.88; 95% CI, 0.86–0.90 at 5 years). Calibration was adequate in North American cohorts, but the original risk equations overestimated risk in some non-North American cohorts. However, through use of a simple calibration factor for non-North American populations, which lowered the baseline risk by 32.9% at 2 years and 16.5% at 5 years, calibration was improved in 12 of 15 and 10 of 13 non-North American cohorts at 2 and 5 years, respectively ($P = .04$ and $P = .02$). This indicated that the KFREs were consistently accurate in predicting progression to ESKD throughout each of the 35 cohorts with discrimination being independent of age, sex, race, or diabetic status. Moreover, the simplified four-variable equation produced similar discrimination to the eight-variable equation (at 2 years C-statistics were 0.89 vs. 0.90 and at 5 years 0.86 vs. 0.88, respectively). An online risk calculator has been produced that utilizes the four-variable equation and incorporates the correction for persons outside of North America (http://kidneyfailurerisk.com/). These findings were a strong indicator that the KFREs would be ready for clinical implementation in the near future.

> ***Clinical Relevance***
>
> **Kidney Failure Risk Equations**
>
> Patients with chronic kidney disease (CKD) are at risk of kidney failure, cardiovascular events, and early mortality. In moderate to severe CKD, estimated glomerular filtration rate (eGFR) and albuminuria are important functional, diagnostic, and prognostic biomarkers, and can be combined in risk equations to estimate the risk of adverse events. The kidney failure risk equations have been developed using eGFR, albuminuria, and readily available demographic and laboratory variables, and predict the risk of kidney failure requiring dialysis or transplant with accuracy. The equations can be used to determine need for nephrology referral, intensity of care, timing of dialysis education, access formation, and preemptive transplantation. Studies evaluating the use of the risk equation in clinical settings are needed.

MODELS PREDICTING DEATH AND CARDIOVASCULAR DISEASE

As mentioned before, patients with CKD progression also evidence an increased risk of death and CVD as they carry a burden of both traditional and nontraditional risk factors for CVD. Existing risk scores have shown poor prediction model metrics in patients with CKD, with no successful attempts in development of a clinically useful model for determining risk of CVD.[328] However one such prediction model developed by Bansal et al. showed promise in assessing 5-year risk of mortality. The risk prediction model, which was developed in a cohort of 828 older adults (mean age 80 ± 5.6 years) from the CHS, included the predictor variables of age, sex, race, eGFR, urine ACR, smoking, DM, and history of heart failure and stroke (C-statistic = 0.72; 95% CI, 0.68 to 0.74). The model was externally validated in 789 participants from the Health, Aging and Body Composition Study (C-statistic = 0.69; 95% CI, 0.64 to 0.74). However, patients with advanced CKD (stages 4–5) in the CHS were underrepresented, and hence the findings would require further validation before clinical implementation. Nevertheless, this model provides a very important starting point for future models to be developed in prediction of all-cause mortality in patients with CKD.[329]

MODELS PREDICTING CHRONIC KIDNEY DISEASE AFTER ACUTE KIDNEY INJURY

A growing appreciation of the risk of CKD and progression to ESKD following an episode of AKI has prompted efforts to develop a risk score for identifying those at highest risk. Three risk prediction models were developed in one study population of 5351 predominantly male veterans with a primary admission diagnosis of AKI to predict the risk of progression to CKD stage 4. Risk factors that entered the models included increased age, low serum albumin, diabetes, lower baseline eGFR, higher mean serum creatinine levels during hospitalization, and severity of AKI as assessed by the RIFLE score (**r**isk, **i**njury, **f**ailure, **l**oss, and **e**nd-stage kidney disease) or need for dialysis, non–African-American ethnicity, and time at risk. The AUROC value was 0.77 to 0.82 for the three models. Sensitivity at the optimal cut-point was 0.71 to 0.77, and specificity was 0.64 to 0.74. External validation was performed on a control population of 11,589 patients admitted for pneumonia or myocardial infarction and yielded good prediction accuracy (AUROC value = 0.81 to 0.82). Sensitivity at the optimal cut-point was 0.66 to 0.71, and specificity was 0.61 to 0.70).[330]

Further validation in other populations that include a more representative proportion of women is required before these risk models can be applied. A validated risk score for patients recovering from AKI may prove very important for identifying high-risk patients for closer follow-up and interventions to reduce the risk of progressive CKD, although further trials are required to evaluate the impact of renoprotective interventions in this setting.

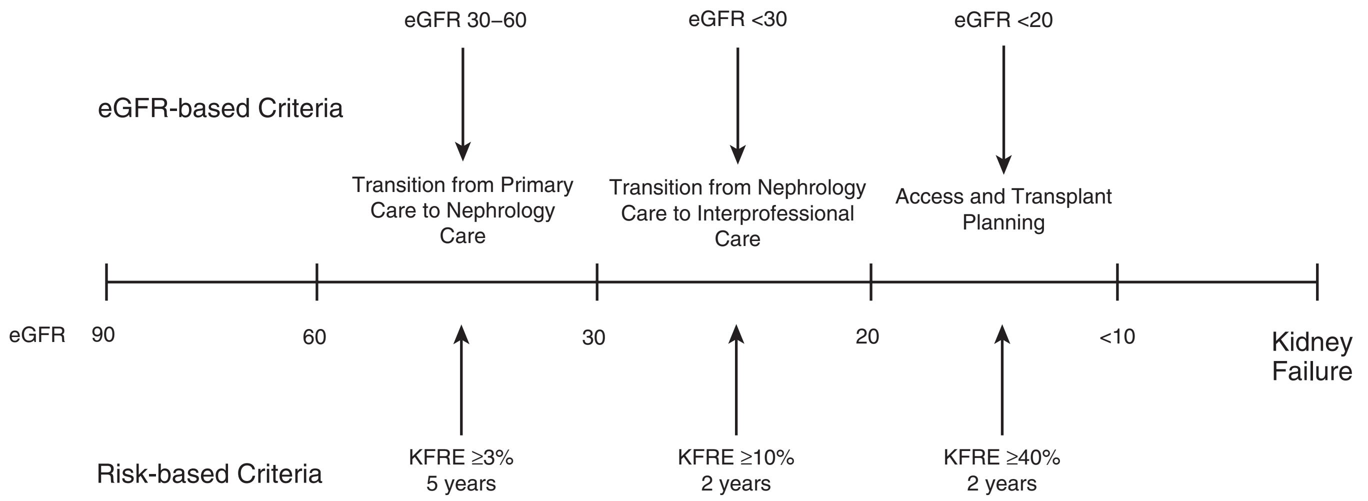

Fig. 20.7 A risk-based versus estimated glomerular filtration rate (eGFR)-based approach to clinical decision-making in patients with chronic kidney disease (CKD). *KFRE,* Kidney failure risk equation. (Reproduced with permission from Tangri N, Ferguson T, Komenda P. Pro: risk scores for chronic kidney disease progression are robust, powerful and ready for implementation. *Nephrol Dial Transplant.* 2017;32:748–751.)

CLINICAL IMPLEMENTATION OF RISK MODELS

Clinical implementation of risk-based care requires the interaction of both a validated risk prediction model and absolute risk thresholds to guide therapy decision. Through the simultaneous use of risk equations and previously set thresholds, an optimal and cost-effective management plan can be provided to the patient. This is currently exemplified by an approach using the KFREs with risk thresholds to prompt transition from one stage of care to the next (Fig. 20.7).[64,331]

PRIMARY CARE TO NEPHROLOGY TRANSITION

In the initial stages of CKD, patients often develop salt and water retention, hypertension, and anemia.[332] Primary care physicians are able to manage these symptoms and provide optimal care for these patients, avoiding both unnecessary spending and treatments. However, when patients exhibit a high risk of CKD progression, referral to a nephrology service and subsequent access to interdisciplinary care teams is necessary in to order to delay CKD progression, prevent suboptimal dialysis initiation, and reduce the chances of early mortality.[6,333] The use of risk equations and risk thresholds may allow physicians to be better informed about when such a referral should be considered. This approach may also prevent low-risk CKD patients from exposure to unnecessary treatments and the anxiety associated with referral.[334] Integrating risk prediction models into clinical care can also be potentially useful in the planning of follow-up. Patients with a high risk of CKD progression but only moderately reduced eGFR can be followed at 3- or 6-month intervals, whereas those with lower risk can be assessed on an annual basis.[335]

INTERDISCIPLINARY CARE

The use of an interdisciplinary nephrology team, which involves a nurse, pharmacist, and dietician, may improve concordance with treatment, reduce suboptimal dialysis initiation, and could even reduce mortality for high-risk CKD patients. As with most specialized clinical services, inclusion of too many patients with a low risk of CKD progression can saturate interdisciplinary resources with minimal benefits.[331] This was a major issue in Ontario, Canada, where health-care administrators struggled with excess demand for interdisciplinary care with a limited health-care budget. As a result, in 2016 the Ontario Renal Network, which is responsible for administering the provinces renal health resources, adopted a risk-based cutoff for eligibility of clinical resources. Patients with a KFRE-calculated 2-year risk of ESRD exceeding 10% or an eGFR < 15 mL/min/1.73 m^2 were eligible for interdisciplinary care. This drastically lowered the number of patients eligible for interdisciplinary care as the previous threshold for such treatment was an eGFR < 30 mL/min/1.73 m^2, irrespective of risk. It was determined using the KFREs that only two-thirds of patients that fit this previous criteria had a 2-year risk >10%. Also from implementation of these new criteria, a small proportion of patients with an eGFR between 30 and 45 mL/min/1.73 m^2 were determined at very high risk of CKD progression and were now eligible for interdisciplinary care. This reallocation of resources led to cost savings of approximately 6 million Canadian dollars annually and has allowed high-risk patients with an eGFR > 30 mL/min/1.73 m^2 earlier access to interdisciplinary care.[336]

DIALYSIS ACCESS AND MODALITY PLANNING

Current clinical practice guidelines recommend dialysis modality planning for patients with an eGFR < 30 mL/

min/1.73 m^2. However the majority of these patients, particularly the elderly, may never actually progress to ESKD.[337] For these patients, the prospect of dialysis creates unnecessary anxiety, and results in unnecessary treatment planning. In particular, the risk–benefit balance for arteriovenous fistula (AVF) insertion is important. While AVF is the standard form of dialysis access for patients on hemodialysis, the benefits may be reduced in the elderly. These individuals suffer from a higher likelihood of primary AVF failure and adverse events including heart failure and steal syndrome. Older people at low risk of developing ESRD also tend to have a higher risk of competing morbidities, thereby making preemptive AVF formation both futile and costly.[338] The use of a risk prediction model to guide the timing of AVF formation can overcome some of these limitations and provide more optimal treatment plans. Patients with low-risk CKD progression will adopt a monitoring-based approach, whereas those with the highest risk of progression can be recommended for access placement for dialysis. In 2014, a Markov model was developed for assessing the implementation of the KFREs on simulated CKD progression in the Sunnybrook hospital cohort. The goal was to determine if the KFREs could select, with the highest accuracy, those patents in need of access for dialysis, while simultaneously avoiding creation of access in patents who either died before ESRD or who did not progress to dialysis within 2 years of access insertion. It was found the KFRE-based threshold of 20% annual risk was superior to eGFR-based thresholds in predicting the need for dialysis and AVF formation.[339]

FUTURE CONSIDERATIONS

Considerable progress has been made since 2000 in identifying risk factors that predict the progression of CKD in diverse cohorts from the general population and nephrology clinics. There is some variation among studies, likely due to differences in the populations and variables studied. Remarkably, a relatively small group of risk factors appears common to many studies, and much progress has been made in developing risk scores based on these variables to predict CKD progression.

Future studies will likely focus on the use of novel biomarkers and genetic factors as risk factors (see Chapter 27) and variables in risk scores, although measurement of such markers is likely to be associated with greater cost than the simple risk factors used to date. Accurate risk scores can align risk and resources while allowing for the most personalized care for persons with CKD. As a result, now is the time to implement these risk scores into clinical care to help patients and clinicians navigate the transition from CKD to dialysis. Further studies are also required to develop risk scores to predict cardiovascular risk in patients with CKD that take into account the close association between CKD and CVD.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. Which of the following is FALSE regarding risk factors for chronic kidney disease (CKD) progression?
   a. More rapid declines in estimated glomerular filtration rate (eGFR) are common in younger patients with CKD, and lead to a higher risk of kidney failure and death
   b. Kidney failure requiring renal replacement therapy is more common in women than men
   c. Ethnicity is a strong indicator of CKD progression to end-stage kidney disease (ESKD), and African Americans are at higher risk
   d. Coding variants in the *APOL1* gene are associated with an increased risk for development of ESKD

   Answer: b

   Rationale: CKD is associated with at least the same relative increase in risk of ESKD in women as in men.

2. Which of the following biomarkers may NOT be useful in risk stratification of patients with chronic kidney disease (CKD)?
   a. Asymmetric dimethylarginine (ADMA)
   b. Neutrophil gelatinase-associated lipocalin (NGAL)
   c. Alpha-fetoprotein (AFP)
   d. Uromodulin (UMOD)

   Answer: c

   Rationale: ADMA, NGAL, and UMOD have all been shown to be useful biomarkers in the risk stratification of patients with CKD.

3. Which of the following is accurate with regard to the relationship between cardiovascular disease and chronic kidney disease (CKD)?
   a. There is only a weak association between CKD progression and death due to cardiovascular disease
   b. Diuretics are contraindicated in treatment of sodium retention–induced hypertension for patients with CKD
   c. Current cardiovascular risk prediction models require further validation in persons with CKD, particularly for those with advanced CKD (stages 4–5)
   d. All of the above

   Answer: c

   Rationale: Current risk scores for CVD have shown poor prediction model metrics in patients with CKD.

4. The kidney failure risk equations (KFREs) use which of the following variables when predicting the development of end-stage kidney disease (ESKD) in patients with chronic kidney disease (CKD)?
   a. Estimated glomerular filtration rate (eGFR)
   b. Albuminuria
   c. Age
   d. Sex
   e. All of the above

   Answer: e

   Rationale: The KFRE uses eGFR + Albuminuria + Age + Sex, to predict ESKD in patients with CKD.